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Damien Le Quéré

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Conception et étude des performances d'une solution auto-configurable pour les réseaux de transport du futur

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devant le jury composé de :

Hacene FOUCHAL
Professeur HDR, Université de Reims / rapporteur

Pascal LORENZ
Professeur, Université de Haute-Alsace / rapporteur

Jean-Marie BONNIN
Directeur de recherche, TB / examinateur

Patricia LAYEC
Ingénieur R&D, Alcatel-Lucent / examinateur

Erwan LE MERRER
Ingénieur R&D, Technicolor / examinateur

Christophe BETOULE
Ingénieur R&D, Orange Labs / examinateur

Adlen KSENTINI
Professeur HDR, Université Rennes 1 / directeur

Bruno SERICOLA
Directeur de recherche, INRIA / co-directeur

I want to dedicate this work to my loving parents and sisters.

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Abstract

In this thesis, we study the LOCARN solution “Low Opex & Capex Architecture for Resilient Networks". LOCARN is an alternative packet network architecture that has been conceived with a special attention to the simplicity of its structure and mechanisms while allowing by design the resiliency and the self-adaptation of clients transportation services. Considering the growing complexification of operators transport networks in recent decades, we consider these latter as the privileged use case. In such a context, LOCARN would allow a drastical simplification of devices and their operation compared to common operator solutions – this involves respectively reductions of CAPEX and OPEX. In this work, we first present LOCARN technically and we bring out its interests for operators beside other transport technologies. Then, since the primary issue of LOCARN is it scalability for large networks, we study this point in details which allow us to establish that the architecture is altogether capable to scale in realistics transport networks. Moreover, to increase the performances we specify two design improvements allowing the architecture to transport a huge amount of clients, the obtained results are very encouraging.

Résumé

Dans cette thèse, nous étudions la solution LOCARN : “Low Opex & Capex Architecture for Resilient Networks ". LOCARN est une architecture de réseaux paquet alternative conçue dans une optique de simplicité de sa structure et ses mécanismes tout en permettant par sa conception la résilience et l’auto-adaptation des services de transport clients. Compte tenu de la complexification croissante des réseaux de transport opérateurs ces dernières décennies, nous prenons ces réseaux comme cas d’usage privilégié. Dans ce cadre, LOCARN permet une simplification considérable des composants et de leur gestion en comparaison des solutions actuelles des opérateurs – ce qui implique respectivement des réductions de CAPEX et d’OPEX. Dans le travail qui suit, nous présentons LOCARN techniquement et mettons en évidence ses intérêts pour les opérateurs par rapport aux autres technologies de transport. Puis, la question prioritaire étant la capacité de mise à l’échelle de LOCARN pour des réseaux de grandes dimensions, nous étudions cette question en détails ce qui nous permet d’établir que l’architecture est tout à fait capable de passer à l’échelle dans des réseaux de transport réalistes. Pour améliorer les performances nous avons également spécifié et évalué deux améliorations de conception permettant à l’architecture de transporter d’un très grand nombre de services, les résultats obtenus sont très encourageants.

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Glossary of Accronyms

ACL Autonomic Control Loop. 27

ANMS Autonomic Network Management System. 50

AODV Ad-hoc On-demand Distance Vector. 2

AP Access Port. 35, 36, 52

APF Adaptive Probabilistic Flooding. 43

ASON Automatically Switched Optical Network. 23

ASTN Automatically Switched Transport Network. 23

ATM Asynchronous Transfer Mode. 5, 7, 11, 13, 20, 31, 32

BFD Bidirectional Forwarding Detection. 45, 46

CAPEX CAPital EXpenditure. 2, 40, 46

CDRP Connection Destination Reference Point. 37, 51

CL ConnectionLess. 12

CL-PS ConnectionLess Packets Switching. 13, 17

CMM Circuit Mode Multiplexing. 18

CO Connection Oriented. 7, 12, 13

CO-PS Connection Oriented Packets Switching. 13, 17

CORP Connection Origin Reference Point. 35, 36, 38, 51, 74

CS Circuit Switching. 13, 17, 18, 26

CU Control Unit. 51

DiffServ Differentiated Services. 20

DSR Dynamic Source Routing. 2, 42, 43

DV Distance-Vector. 32, 33, 42, 46

DWDM Dense WDM. 18

EMS Element Management System. 22

EN Edges Node. 35–37, 51, 76

EU Edge Unit. 51

FDM Frequency-Division Multiplexing. 18

FEC Forward error correction. 16

FIB Forwarding Information Base. 36, 40

FO Fiber Optic. 6, 21, 46

FRR Fast ReRoute. 24, 45, 46

FTTH Fiber To The Home. 5

GMPLS Generalized Multi-protocol Label Switching. 1, 23–25, 29–32, 45

HLO High Level Objective. 28, 32, 50

IGMP Internet Group Management Protocol. 76

IGP Interior Gateway Protocol. 32, 45, 46

IntServ Integrated Services. 20

IP Internet Protocol. 1, 11–13, 17, 20, 21, 31, 32, 44, 45, 48, 76, 85

IS-IS Intermediate System to Intermediate System. 23, 32, 45

ISIS-TE ISIS Extensions for Traffic Engineering. 32, 45, 46

ITU International Telecommunication Union. 6–9, 11, 13, 16, 18, 20, 21, 23, 35, 82

LDP Label Distribution Protocol. 45, 46

LMP Link Management Protocol. 24

LS Link-State. 32, 33, 42, 46

LSP Label Switch Path. 24, 25, 32, 46

LSR Label Switch Router. 24

MAN Metropolitan Area Network. 5

MAPE-K Monitor Analyse Plan Execute Knowledge. 27

MEF Metro Ethernet Forum. 7, 13, 15, 21

MPLS Multi-protocol Label Switching. 1, 11, 13, 20, 23, 25, 26, 31–33, 41, 44–46, 48, 85

MPLS-TP Multiple Label Switching Transport Profile. 6, 13, 15, 20, 21, 31, 32

NE Network Element. 22, 23, 30

NMS Network Management System. 22, 23, 30, 45, 50

OAM Operation And Maintenance. 15, 21, 31, 38, 41, 42, 45, 49, 51, 57, 58, 60–62, 73–76, 80–82

OI Optimization Interval. 39, 49, 60, 61, 69, 79

OLSR Optimized Link State Routing. 2

ONF Open Networking Foundation. 29

OPEX OPerational EXpenditure. 2, 40, 46

OSPF Open Shortest Path First. 23, 32, 45

OSS Operations Support System. 50

OTN Optical Transport Network. 5, 13, 19, 21, 44

P2MP Point-to-MultiPoint. 76, 77, 82, 83, 86

PB Provider Bridges. 21

PBB Provider Backbone Bridge. 21, 44

PBB-TE Provider Backbone Bridge Traffic Engineering. 6, 21, 31, 32, 44

PCE Path Computation Element. 24–26, 29, 31, 50

PDH Plesiochronous Digital Hierarchy. 18, 19

PDU Packet Data Unit. 19

PIM Protocol Independent Multicast. 76

PS Packets Switching. 2, 7

PTN Packet Based Transport Network. 20, 31, 44, 85

QoS Quality of Service. 14, 19, 20, 31, 49, 62, 69, 86

RP Reference Point. 35

RSVP Resource reSerVation Protocol. 24, 25

RSVP-TE RSVP Extensions for Traffic Engineering. 23, 45, 46

RTT Round Trip Time. 37, 38

RU Relay Unit. 51

SCI Service Check Interval. 49, 60, 61, 82

SDH Synchronous Digital Hierarchy. 5, 7, 11, 13, 14, 19–21, 44

SDN Software Defined Networking. 29–31, 33, 50, 86, 87

SLA Service Level Agreement. 6, 14

SLS Service Level Specification. 6, 14, 51

SON Self-Organizing Network. 2, 32

SONET Synchronous Optical NETwork. 5, 13, 14, 19–21, 44

SPB Shortest Path Bridging. 32, 45

sTDM statistical Time-Division Multiplexing. 19

STP Spanning Tree Protocol. 21

TDM Time-Division Multiplexing. 18, 44

TE Traffic Engineering. 32

TN Transit Node. 35, 37, 51, 76

TRILL Transparent Interconnection of Lots of Links. 32, 45

TTL Time To Live. 37, 39, 41, 42, 49, 53, 55–62, 67, 69, 70, 79, 80

UCP Unified Control Plane. 23

VLAN Virtual Local Area Nework. 21, 44, 48, 49

WDM Wavelength Division Multiplexing. 6, 18, 21, 24

Chapter 1

Introduction

1.1 Motivations

Analysts expect a strong evolution of the types of services carried by network providers in the future. New services with high added value such as bandwidth on demand, “cloud computing" storage networks for example, require more automation. These new uses are transforming the concept of “permanent connection" into the concept of “on demand" connection which require network resources settling time and release to be adapted to the carried services. Most of current network operators systems rely on a transmission optical physical infrastructure. Although it is generally fixed and stable, there are other instances of media and architectures that face changing physical characteristics (microwave, Wi-Fi...). This is particularly noticeable in the case of transmission networks to build in difficult and unstable environments. For example, energy availability may be random, resources can be subject to very long periods of downtime, and the deployment should be achieved before the results of planning studies. Other external factors may also be considered. Note for example the sensitivity of the environment to social and political contexts, the use of infrastructure provided by a third party which the performances are not under control – such contexts can match those of emerging countries. All this leads operators to consider more flexible and dynamic transport networks.

To meet these needs, several tracks are under study. In the short term, manufacturers offer to enrich existing networks by adding functional components and new protocols that complicate them meanwhile they do not guarantee that the performance obtained are consistent with the needs – see Multi-protocol Label Switching (MPLS) extensions for the Internet Protocol (IP) and the Generalized Multi-protocol Label Switching (GMPLS) control plane

applied to optical networks.

Other approaches aim to improve the self-organizing properties of networks. Some are based on autonomic principles like for example Self-Organizing Networks (SONs), others rely on specific overlay layers. Some others, based on protocols such as Dynamic Source Routing (DSR), Optimized Link State Routing (OLSR) or Ad-hoc On-demand Distance Vector (AODV), are particularly studied for the wireless ad-hoc networks because of their needs for adaptations.

LOCARN (Low Opex Capex Architecture for Resilient Networks) have been though in the framework of the European CELTIC Tiger2 project [3] in 2010. It is a Packets Switching (PS) architecture that removes all information states and routing processes in transit nodes, and offers a plug&play self-adaptive network. LOCARN is characterized by a transfer scheme using autoforwarding and a control plane based source routing discovery mechanism using network floodings. The information collected by the control plane is used to populate the required fields of the autoforwarding client frames. LOCARN is well suited for a very simple establishment of point-to-point connections and maximization of network resources use – which allows to reduce the OPerational EXpenditure (OPEX). Furthermore, it involves basic devices that are very simple to implement – which allows to reduce the CAPital EXpenditure (CAPEX).

All these emerging solutions provide opportunities for operators to rethink the architecture of their transport networks by optimizing the interaction between the services and the infrastructure that transport them (mainly based on fiber optic today).

1.2 Contributions of the Thesis

At the begining of the thesis, LOCARN was (i) an architecture idea envisioned for transport networks (ii) a proof of concept plateforme used for the communication with both the scientific community and internally to the Orange group. With this thesis, the LOCARN architecture has been positioned, evaluated and improved which results through several contributions:

– *A state of the art about Transport Networks* has been done to position LOCARN among network operators technologies, in particular regarding metro/core networks.

It includes an analysis of the standards and of transport networks historical evolution to bring out their perspectives.
- *The positioning of LOCARN* beside its conceptual ancestors (DSR and APLASIA) and the closest operator alternative (IP/MPLS)
- *A simulator of LOCARN has been entirely developped* upon the OMNet++ framework
- *An extensive study of LOCARN performances and ability to scale* through discrete event simulations and analytic modeling, focusing particularly on the control plane. Those results of have been published in an international conference [56] and received the “best paper award prize".
- The previous work has been extended by a probabilistic modeling to bring out the expected impact of network failures on the architecture. The overall LOCARN study has been published in a journal special issue [57].
- *Two improvement proposals have been specified*, evaluated by simulation and presented at the Globecom 2014 international workshop [55]. The results of those improvements are very encouraging concerning the architecture scalability. In addition, the second improvement allows LOCARN to provide point-to-multipoint communications without the involvement of an additional protocol – on contrary to widespread multicast solutions (e.g. PIM/IGMP) – while keeping all its good properties, namely simplicity, self-configuration and resiliency.
- Another study comparing the control plane performances with OSPFv2 has been achieved and presented internally at Orange during the “journée des doctorants" in the form of a poster. The paper “LOCARN: An innovative Plug&Play auto-adaptative packet transport network" is still pending.

1.3 Organization of the Manuscript

In chapter 2, the state of the art is developed for a sound positioning of LOCARN. This state of the art reviews operators Transport Networks since they constitute the envisioned use case for LOCARN in this thesis. This chapter aims to expose both a theoretical and a practical view of those networks in order to bring out the key factors of their evolution and the current trends. Finally, the benefits of the LOCARN application to Transport Networks are brought out. In chapter 3, the LOCARN architecture is technically presented as follows: (i) how it works ; (ii) its general pros and cons ; (iii) its positioning besides its two conceptual ancestors ; (iv) its positioning as a transport technology ; (v) its positioning as a control and management proposal ; (vi) information related to its modeling and simulation. In chapter 4,

the performance and the scalability of LOCARN are evaluated. First into a “normal" context where the overhead cost related to the control plane is estimated analytically and confirmed by simulation. In a second step, the LOCARN performances are assessed in probabilistic terms in the case of failures (of links or node) using a Markov chain model whereas the impact of failures is estimated by simulation. In chapter 5, two improvements are proposed to constitute a "Large Scale" version of the LOCARN architecture. Both the two proposals are described and evaluated by simulation, their perspectives are briefly discussed. Finally, in chapter 6 we conclude about the thesis contributions and results.

Chapter 2

Transport Networks: Definition, Evolution and Automation Perspectives

2.1 Transport Networks: Definition

A telecommunication network is a complex network that can be described in a number of different ways depending on the particular purpose of the description. The “Transport Network" terms are widely used among a lot of Information Technology’s areas under many different meanings. Thereafter in this chapter, we focus on the Network Operators viewpoint and usage of the “Transport Network" terms. At first glance, it is even not possible to provide a global definition for operators, due to the multiplicity of usages. Yet it is possible to identify four viewpoints:

1) The operator *architectural viewpoint* Based on an historical distinction between an “Access Network" as opposed to a “Transport Network". This viewpoint – inherited from telephone networks – is used as a convenient appellation that reflects some technological ruptures in operators architectures (for physical layers as well as for upper ones and the associated control/management approaches). Yet a such transport network delimitation becomes more and more questionable: do Metropolitan Area Networks (MANs) belong to Transport Networks ? Does the Transport Network start at the customer by considering the Fiber To The Home (FTTH) deployments ?

2) The operator *“de facto" viewpoint* It uses the Transport Network appellation as an umbrella gathering the technologies classified as transport ones. In the historical order, the designated technologies are: Synchronous Digital Hierarchy (SDH) and Synchronous Optical NETwork (SONET), Asynchronous Transfer Mode (ATM), Optical

Transport Network (OTN) and Wavelength Division Multiplexing (WDM) for circuit technologies ; more recently for the packet ones: Multiple Label Switching Transport Profile (MPLS-TP) and “Carrier Ethernet" technologies in particular Provider Backbone Bridge Traffic Engineering (PBB-TE). In practice, it is the most widely accepted definition because it is both the most convenient and accurate one. To have a an overview of the technical specifications of transport technologies, a significant comparative work has been done in 2010 for the european GEANT network [11]. The latter uses the terms of “Carrier Class Transport Network Technologies".
However a such viewpoint remains arbitrary since no general attributes are exhibited in order to provide a fundamental Transport Networks' definition.

3) The *equipments vendors viewpoint* It is focused on equipments characteristics: the transmission error rate, devices reliability, or the distance of transmission. Most of time, this point of view leads to make “Transport Networks" synonymous with the Fiber Optics (FOs) because of the transmission properties they confer to infrastructures. Yet, even if this viewpoint is significant, it solely provides a partial definition.

4) The *clients viewpoint* For a client, a transport network simply designates an underlying system seen as a black box and achieving transportation between somme access points. Here the transportation process is considered as a service that must be mastered and characterized in order to be ultimately contractualized under Service Level Specifications (SLSs) and Service Level Agreements (SLAs) documents. This definition is also meaningful but it remains partial since it says nothing about the technical characteristics of transport networks.

We argue that all those viewpoints allow to understand the operators considerations about their “Transport Network". By assuming all of them, we finally bring out that two orthogonal aspects must be considered: the *transport network functional aspect* and the *transport networks operational aspect*. Those can be defined as follow.

Transport Networks Functional Aspect Fundamentally, a “Transport Network" is a network considered from the information transfer capability viewpoint. This is the definition of the *transport functional group* used in G.8XX recommendations [6, 9, 13] from International Telecommunication Union (ITU). Under the ITU-T definition, the *transport function group* designates the set of functional components “which transfers any telecommunications information from one point to another point(s)"; it is opposed to the *control functional group* which “realizes various ancillary services and operations and maintenance functions". Thus,

in a functionnal acceptation a transport network is reduced to the first group. Such a functional definition is finally unrestrictive; it is to say that functionally speaking any telecommunication networks is a transport one and can be depicted by using G.8XX definitions and notations.

Originally, the model has been proposed in 2000 (through the G.805 [6] recommandation) in order to constitute an harmonized set of recommendations for the depiction of Connection Oriented (CO) networks under a single notation, namely in those days SDH and ATM. This functionnal model has been extended in 2003 to PS technologies with G.809 [9], and finally unified in 2012 with G.800 [13]. Nevertheless, all the fundamental concepts of the functional model are already exposed in G.805.

Transport Networks Operational Aspect In practice, besides the functional definition, operators use the “Transport Network" terms in a more restrictive meaning. Indeed, for an operator, a Transport Network is not simply a network capable of conveying information, but it is a network that has been conceived, built and that is maintained by considering that transportation is *a service provided to clients*. By extension, the transportation service(s) must be technically characterized, contractualized. Morevover, this leads operators to discriminate network solutions between transport network technologies and others ones. This discrimination is done according to solutions ability to reach some technical requirements like performances or scalability for example. It must be noted that, finally, this discrimination does not exclusively depends on the transport functional group cited above.

Usually, the Transport Networks technical characteristics are refered as “Carrier Grade" requirements and properties. A set of requirements have been clearly expicited by the Metro Ethernet Forum (MEF) in order to promote the Carrier Ethernet technologies. This set of requirements, used by the MEF as a touchstone for certifications, are helpful to bring out the operators technical chalenges of metro-core networks. Nonetheless the Carrier Grade properties must be actually understood as reflecting the concrete (technical or administrative) issues that the operators have to deal with but they are not absolutely fixed rules. It means that fundamental changes appearing in operators context or needs would lead to refine or even reconsider those rules, and thus the “Transport Network" terms effective definition.

In this section we define the Operators’ Transport Networks in order to understand their evolution in Network Operators contexts. To do so, we first overview the functionnal definition exposed in ITU-T model which allows to depict the *transport functional group* (subsection 2.1.1). Then, we overview the specificities of operators transport networks accepta-

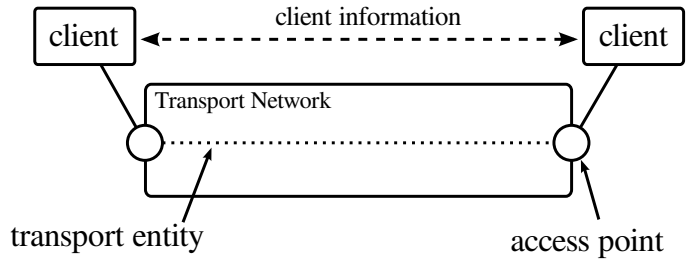


Figure 2.1 Client representation of a transport network

tion by outlining the *operational properties and requirements* (sub-section 2.1.2).

2.1.1 Functional View

The ITU-T Functional Model (G.805)

First step: the client viewpoint At first, a transport network can be seen from a client viewpoint. Following the client representation, a transport network is a system providing some services of information transportation between a set of locations (see Fig. 2.1). Thereafter, the terms “access point" are often prefered to the “location" one, whereas a “Service" designates a set of two or more access points and their associated transport entities for a specific client purpose. The ITU-T transport functional model extends and generalize this client/server view of the information transportation by providing architectural components and rules for the depiction of all the possible interconnections through the transport functional group. In other words, to depict internally the overall “Transport Network" that is a black box in the client representation (see Fig 2.1).

In the ITU-T model, the network can be decomposed into generic architectural components defined by the role they play in the information transport process. The model also profides a set of attachment rules defining which components can be interconnected. The model defines four kinds of architectural components: (i) *topological components*: *network layers*, *subnetworks*, *links* and *access groups* ; (ii) *transport entities*: *connections* and *trails* ; (iii) *transport processing functions*: *termination functions* and *adaptation functions* ; (iv) *reference points*: *access points* and *connection points*. Excepting topological components that provide the more abstract description of a network, all components can be unidirectional or bidirectionnal.

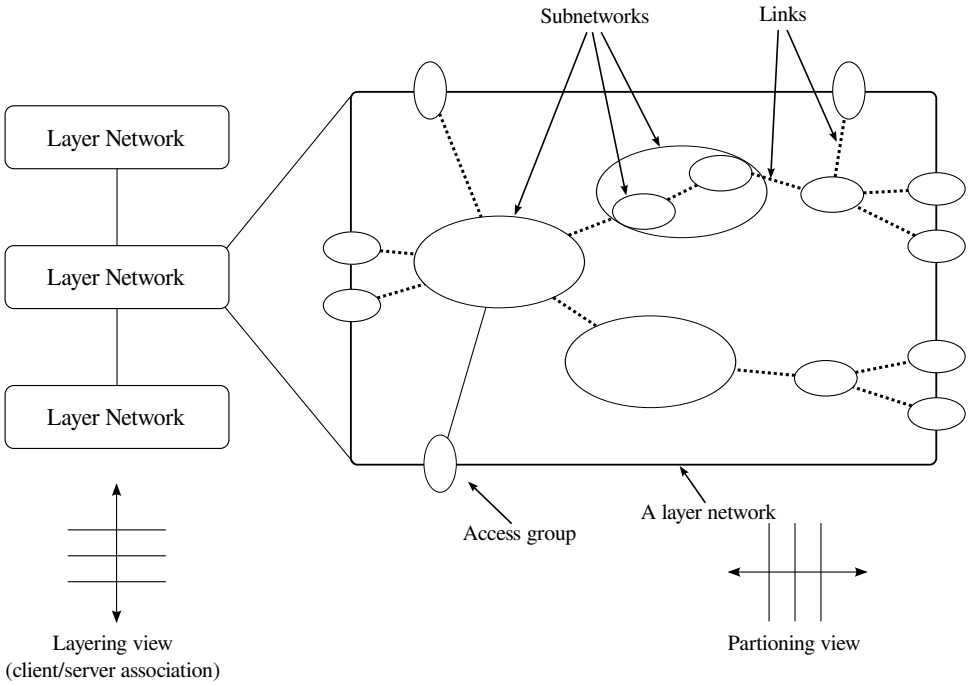


Figure 2.2 The layering and partitioning principles (G.805 notation)

Fundamental decomposition principles The two important concepts of the G.805 architecture are recursive principles of networks decomposition called *layering* and *partitioning*. ITU-T's model follows those decomposition principles because the *transport notion is itself recursive*. Indeed, the transport between two locations may be achieved through a direct link, but it can also be achieved by passing through several intermediate locations. In the second case, since a transport process is defined by the information displacement between locations, the initial transport process (called "end-to-end") can be decomposed into several successive transport processes. This decomposability is the basis of transport networks *partitioning* (or "horizontal" decomposability).

In the same manner, considering a given network composed of several layers, what is called the transport process in a specific layer is the information transfered between locations through the reference points of this layer. Yet the transport process through this network layer is achieved by using an underlying transport process of one or maybe several sub-layers. The relationship between two adjacent network layers is called "client/server". This relation is characterized by the fact that the underlying layer (called server) provides to the upper layer (called client) the mean to transport its information between locations such that the client layer does not have to be aware of how it is transmitted between its attachment points through the server layer – this principle called "transparency" is fundamental.

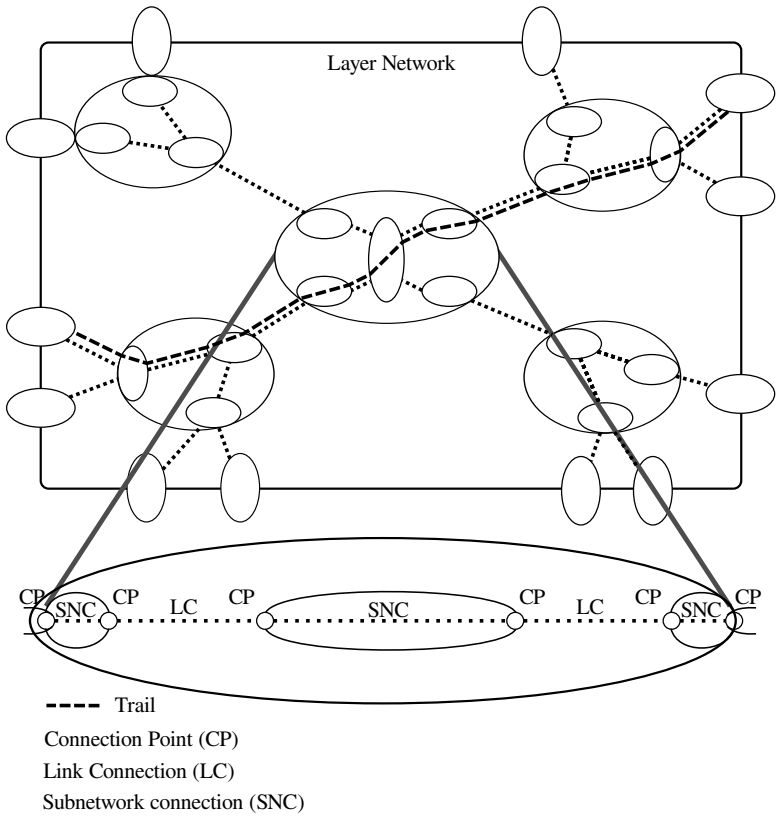


Figure 2.3 Relationship between partitioning of subnetworks and decomposition of connections (G.805 notation)

In the G.805 model, those decomposition match to decomposition principles corresponds to the decomposability of the topological elements. Layering corresponds to the staking of *network layers* whereas partitioning matches with the subdivision of *network layers* into subnetwork and links (see Fig. 2.2).

Architectural components in more details Now let us overview the architectural components and their relationship (see Fig. 2.3). A *network layer* provides to its clients a transport service between *reference points* called *access points*. A *network layer* is defined by a set of *access points* of the same type (i.e. able to communicate together), that are grouped by locations, forming so called *access groups*. *Access points* can be connected together by *transport entities* called *trails* corresponding to a *network connection* (an end-to-end network path) surrounded by *termination functions*. The termination functions are *transport processing functions* that adds or removes information respectively at the input or output of a *network connection*. The added information is called *layer information*, typically it allows the corresponding trail to be monitored (e.g. labelling, sublayer fields). *Connections* are the

transport entities related to the topology of the *network layer*. An end-to-end *connection* through a *network layer* is called a *network connection*. According to the topological decomposition of the *network layer* into *subnetworks* and *links*, a *network connection* can be considered as a succession of *link connections* and *subnetwork connections*, interconnected by *connection points* and terminated by *termination connection points*. This topology decomposition is recursive, as a *subnetwork* can itself be decomposed, until the representation involves only links.

An important point to retain from the G.805 model is that a *network layer*, is designed as the basic building block of a transport architecture. It corresponds to the abstract representation of network technologies such SDH, ATM, IP, Ethernet, MPLS in the model for example. The independence of a *network layer* is the compliance with two principles of the client-server transparent relationship: the topology independence between layers, and the information's form independence between layers. In the architecture, the layering relationship between two adjacent layers corresponds to the attachment of a *client network layer* to the *server's access points*. Exactly, a connection of the *client layer* match with a *trail* in the *server layer*.
In the architecture, the topological independence between layers corresponds to the fact that a *client network layer* only considers the *access points* of the *server network layer* and the *trails* between them. A client connection relies on a *trail* of the *server network layer* whatever this *trail* corresponds to in terms of *server connections*, ie whatever the underlying server topological arrangement ("transparency"). The topological independence of *network layers* results from the structural client/server relationship.

On the other hand, the information's form independence corresponds to the introduction of *transport processing functions*. So far in this ITU overview we talked about "information" transport without more precision. However each network layer uses its own information transmission form, this layer specific form is called *characteristic information*. To put the information into this form, some *transport processing functions* called *adaptation functions* are used between the adjacent network layers. The *adaptation functions transform* the information from the client to the server layer, and vice versa. In the first case it will adapt the *client characteristic information* into a form suitable for transport over a *trail* in the *server network layer* called the *server adapted information*. In the second, it will perform the reverse operation in order to restore the initial client information form (i.e. "transparency" of the information form). Modulation, encoding, fragmentation, encapsulation and their sym-

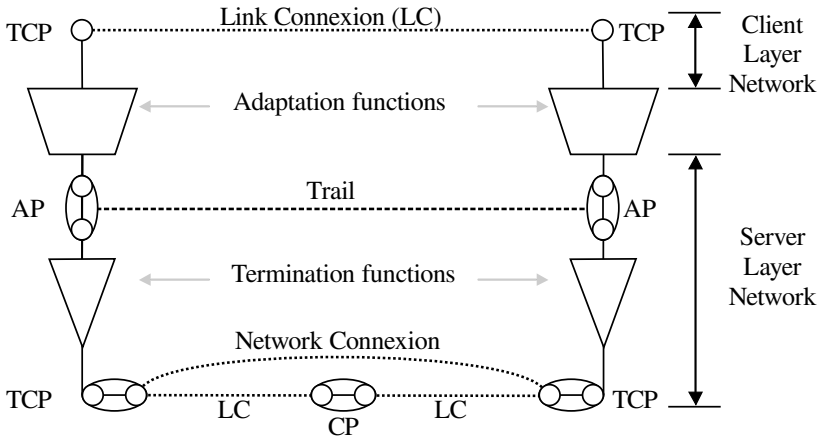


Figure 2.4 Interconnection of two network layers (G.805 notation)

metric operations are some examples of adaptation functions. Since G.800, what is called *characteristic information* of a layer corresponds to the *adapted information* plus the *layer information*. The Fig. 2.4 shows the relationship between two layers in the G.805 notation through an example.

In conclusion, the G.805 model allows to understand operators Transport Networks by depicting their *transport functional group*. On contrary to other network layering models which associate functionnalities to a fixed set of network layers, namely like OSI or TCP/IP, this model is based on the *recursive* decomposition – both vertical and horizontal – of the transport functional components. In this approach, all network layers (i.e. *network layers*) actually have the same function: transport information from one location to another location. Thus, as G.805 is a descriptive model, it does not directly deal with performances issues, efficiency or manageability of the network.

Other network models using G.805 concepts

ITU-T G.809, G.800 The G.809 recommendation [9] (2003) is based on the same decomposition principles than G.805, but it introduces new concepts in order to describe connectionless behaviour and therefore provide a common framework between the CO network layers and a ConnectionLess (CL) network layer. For example, in a connectionsless network layer, a connection become a flow, a trail become a connectionless trail and a subnetwork become a flow domain. The connectionless transport network functionality is described from a network level viewpoint, taking into account network layer structure, networking topology, client characteristic information, client/server layer associations and mapping between CL

and CO network layers. A relevant comparision of general and technical characteristics of SDH, OTN, ATM, Ethernet and IP is provided in annex A of the G.809 recommendation's.

The G.800 recommendation [13] (2012) describes an unified functional architecture for transport networks, Circuit Switching (CS), Connection Oriented Packets Switching (CO-PS) and ConnectionLess Packets Switching (CL-PS) in a technology-independent way by unifying the G.805 and the G.809 models.

Metropolitan Ethernet Forum The Metro Ethernet Forum (MEF) thats aims to promote Carrier Ethernet deployment, publishes a set of technical specifications about various Carrier Ethernet aspects. In the architectural specification [10], a layering model describes metropolitan ethernet networks and includes three layers: application services layer, ethernet services layer and transport services layer (see figure 1). In this model, a given network layer (e.g. IP, MPLS, SONET/SDH) may play a dual role with the Ethernet services layer: (i) as a transport layer providing transport services to the Ethernet service layer ; (ii) as an application service layer using the service provided by the Ethernet service layer. This view corresponds to the ITU's principle of recursive network layering.

The whole MEF architecture is compliant with the G.805 and G.809 recommendations. The architecture is even described with G.805 components in MEF4.

2.1.2 Operational View

As stated before, in an Network Operator viewpoint, Transport Networks are defined by their ability to reach some technical requirements. The emergence of two packet transport technologies (namely Carrier Ethernet and MPLS-TP) has led to exhibit some of those properties. First, the MEF defines five attributes to Carrier Grade Ethernet. In 2009, for the constitution of the MPLS-TP framework [25], then the IETF has provided the MPLS-TP requirements [64].

Metro Ethernet Forum's five attributes

Scalability Network Providers require that the network scale to support the 100,000s of customers to adequately address metropolitan and regional served areas. It must also ensures the support of wide range of interface speeds to support different traffic demands and allow growing the network.

Reliability Protection enables end users to rely on Carrier Ethernet to run their business and mission critical applications. This really implies reliability and resiliency, as service

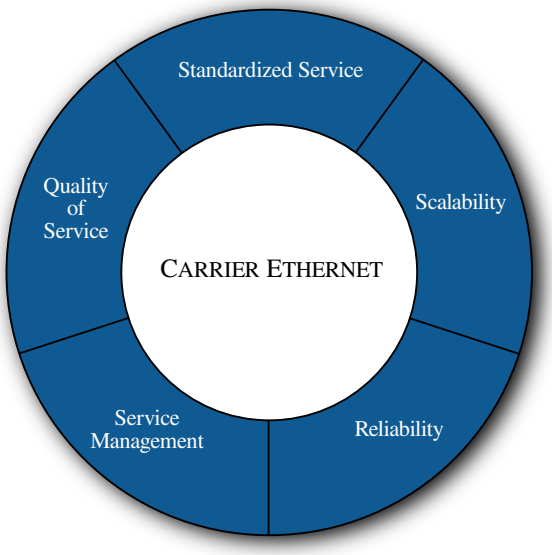


Figure 2.5 The five attributes of Carrier Ethernet (MEF)

providers typically boast 99.999% of network availability. One of the benchmark tools for achieving this has been SONET/SDH's ability to provide 50ms link recovery, as well as protection mechanisms for nodal and end-to-end path failures. For Carrier Ethernet to be adopted – especially in support of converged, real-time applications – it must match these performance levels seen by traditional WAN technologies.

Hard Quality of Service Network providers must be able to offer customers differentiated levels of Quality of Service (QoS) to match application requirements. QoS mechanisms provide the functionality to prioritize different traffic streams, but Hard QoS ensures that service level parameters agreed for each level of service are guaranteed and enforced across the network. This provides customers with the guaranteed, deterministic performance they receive from their existing leased line services. Service providers must also be able to ensure that services receive the performance requirements according to the SLS.

Service Management Network providers require mature network and service management systems that firstly allow quick services provisioning in order to deliver existing and new services and secondly monitoring different parameters of the provided services. Such monitoring is used against an SLA and the service provider must have the performance measurements to back up any service level claims. If a fault occurs, the service provider needs to have the troubleshooting functionality to locate the fault, identify which services have been impacted and react appropriately. The troubleshooting func-

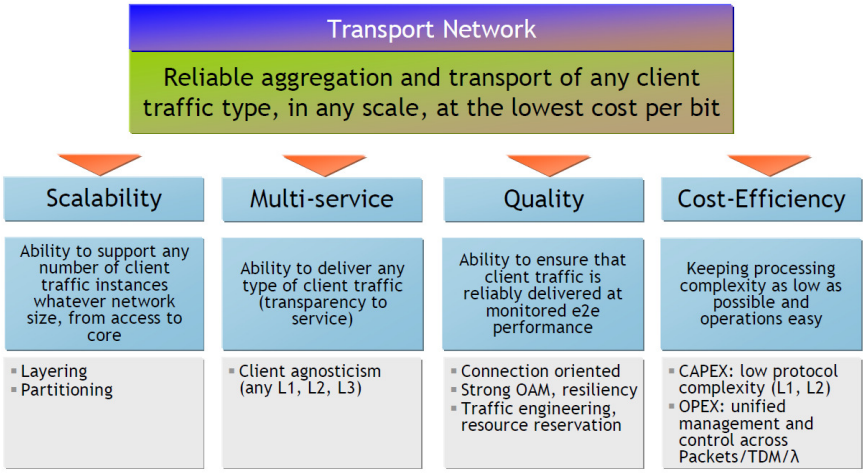


Figure 2.6 MPLS-TP Transport Networks Requirements (source: [22])

tionality also designated by the terms Operation And Maintenance (OAM).

Standardized Services Standardized services enable subscribers, service providers and operators to coordinate in order to achieve data connectivity based on Carrier Ethernet between multiple subsciber sites across multiple operator networks, as required by organizations around the globe.
Using standardized services provides the service provider with the guarantee of interoperability between any network elements that are MEF-certified.

MPLS Transport Profile requirements

For the specification of the MPLS-TP framework [25], the IETF defined a set of requirements [64]. In [22], the MPLS-TP requirements have been summarized through general Transport Networks drivers: (i) *Scalability* (ii) *Multi-service* (iii) *Quality* (iv) *Cost-Efficiency*. See Fig.2.6.

2.2 Transport Networks Historical Evolution

To understand the network operators historical evolution of designs and technologies, it is decisive to understand the continual search for a global *efficiency* of the *transport functional group*. As a general definition, the *transport efficiency* of a network is the ratio from the “transport capacities provided" to the “transport invested resources". Many subdefinitions can be derived according to the considerations of the transport capacities provided and the the transport resources invested. Commonly, studies about the “network efficiency" reduce

the resources investement to the financial cost and the transport capacities to data rates (hence the commonly employed metric is the “Average Per-Bit Delivery Cost").

2.2.1 The Efficiency Driver

In the light of the ITU-T’s G.805 structural model presented above in section 2.1.1, it is possible to locate and categorize the factors of efficiency of the transport functional group as follows.

Efficiency within a network layer

Efficiency of the information processing functions (Adaptation/Termination) First, the efficiency of a network layer is related to the *adaptation functions*, i.e. to the representation of the client information in order to transport it. Indeed, according to the *representation*, a network layer will be more or less efficient. This factor, related to information theory fields, corresponds to many different aspects of the information representation according to the network level: at the physical level it deals with signal issues (e.g. spectral efficiency), above can be found some encoding and compression issues, and above the packetization issues (e.g. packet dimensions, headers and fragmentation).

Then, the efficiency of a network layer is also related to the *termination functions*, i.e. to the amount of non-client information added for different purposes. It is possible to distinguich information added along a Link Connection or along an end-to-end Network Connection, in order to supervise it (e.g connection state, traffic monitoring, Forward error correction (FEC) addition) and the information transmitted by “inband" control/management planes.

Efficiency of topological configuration For a physical network layer, a topology configuration can be considered as more efficient than another one if – for the same client interconnection – it involves less links, less nodes or less transfer capacities of links (according to the specific efficiency aspect considered). For example a star topology involves less links than full mesh but it require one more node if compared to a ring or a bus topology.

For a path network layer, the efficiency is related to the paths configuration. For example, the shortest criteria globally improve efficiency through minimizing the amount of links and nodes used for each connections. On the other hand a network that have

to cope with a sporadic and unbalanced traffic model may be more efficient with an elastic path selection by making a best use of its overall resources (load balancing).

Efficiency of resources allocation / occupancy rate Finally, the efficiency of a network depends on the adequacy between the transport resources it provides and the effective client usage. Hence, it is strongly related to: (i) the adequacy between the kind(s) of services transported and the approach for resources allocation (CS, CO-PS, CL-PS) ; (ii) the adequacy between client effective needs and network sizing: obviously an oversized network is not efficient.

Efficiency of multi-layer networks

Multi-layer arrangement Globally, the superposition of network layers is a factor of inefficiency since it involves additional information processing functions (adaptation, termination) introducing more “non-client" information. Generally, a good point to be efficient is thus to reduce as much as possible the amount of superimposed network layers. Since it leads to unappropriated cross-layer topological arrangements. For example if network layers management is not coordinated, an IP shortest path considering 3 hops may actually correspond to many hops through the underlying infrastructure.

The global efficiency of a multi-network layer lies on the adequacy between the contiguous layer topologies, which most of time leads to complex optimization issues that are addressed by mathematical modeling or multi-objectives constraint optimization programs.

2.2.2 Overview of Transport Technologies

Dedicated Lines versus Transport Networks The most basic way to transport some information between remote clients is to deploy a physical line especially between their location. In its strongest sense, a dedicated line is further specific to an applicative usage (one line for one “service"). The most popular illustration of dedicated line is the Moscow–Washington hotline during the cold war. Based on this observation, the most basic way to design a transport infrastructure would be to deploy dedicated lines between all clients locations, and even one line per applicative usage.
However, if it is the most basic transport design, it is clearly the most inefficient. In a world where infrastructures involve *costs* and have *impacts* – purchase, deployment, maintenance, energy, environment, etc – communications networks can not consists in a such tangle of telecommunication lines. For this reason, it is noticeable that the historical driver of transport networks is ultimatelly the efficiency defined above. To increase networks’ efficiency,

the widely used design principle is the *mutualization of resources*, which can take several forms.

Instead of dedicated lines deployments between final locations, a Transport Network is able to involve an intermediate components used by several end-to-end communications. Those latters perform *switching* between their interfaces in order to establish an end-to-end network trail in the ITU denomination (e.g. a circuit or a route). Hence, switching corresponds to a *mutualization of transmission resources* within a network layer (horizontally). On the other hand, instead of using a physical link for each client communication like the dedicated line, it is possible to *multiplex* several clients communications into a single (server) channel. Hence, multiplexing is the *mutualization of transmission resources* between a client and a server network layer (vertically).

Circuit Switched Transport Networks – "Bellheads" Paradigm

Paradigm Circuit Mode Multiplexing (CMM) consists in the multiple simultaneous transmission of information (streams/signals) on a carrier connection by partitioning the channel bitrate into fixed subchannel bitrates, allowing to transport several client services having *constant bitrates* through a single connection – the level of subdivision's fineness is called "granularity". Hence, CMM is a mean to master the occupancy rate in a CS network and the adequation of the bitrates distribution with the effective clients needs is a decisive factor of efficiency.

In practice there are many types of CMM. In analog transmissions it is common to multiplex the signal according to its physical characteristics. With Frequency-Division Multiplexing (FDM) for example a carrier signal is divided into subchannels having different frequency widths. FDM itself includes optical multiplexing methods like WDM or Dense WDM (DWDM) (wavelenght is the inverse of the frequency). In digital transmission, signals are commonly multiplexed with Time-Division Multiplexing (TDM) where several signals are carried over the same channel by using specifics time slots. At the time when the constant bitrates services constituted the main part of the communications (i.e. telephony), the application of the CMM led to design transport networks under the form of bitrate hierarchies. The circuit switched paradigm comes with a control/management approach where the network operational processes must be mastered and planed in advance.

History In the 60s and 70s, the first deployed digital hierarchies – that are now refered to as Plesiochronous Digital Hierarchys (PDHs) – were based on plesiochronous multiplexing mechanisms, it is to say TDM without synchronization.

In the 80s and 90s, synchronous digital hierarchies – Synchronous Optical NETwork (SONET) [8] in North America and Synchronous Digital Hierarchy (SDH) [5] in Europe – have been developed to overcome the PDH limitations, namely: (i) the lake of visibility of the low bit rates affluents ; (ii) the complexity of the mutliplexing mechanisms due to the frames justification with extra-digits ; (iii) the lake of standaradized management (iv) the lake of bytes dedicated in frames to the operation ; (v) no global interoperability because of the different bitrates between Europe, Japan and North America. Finally, SONET/SDH hierarchies allowed to evolve towards higher bitrates transport networks, to globally interconnect telecommunication hierarchies and to improve transport networks management towards a better flexibility of multiplexing and the extension of OAM features (with a few exceptions, SDH can be thought of as a superset of SONET).

From the 2000s to now, the Optical Transport Networks (OTNs) hierarchy [7] has been conceived to extend SDH and adapt the bitrates to the increasing demand and to provide additional OAM features.

Packet Switched Transport Networks – “Netheads" Paradigm

Paradigm The obvious specificity of a packet network is to put the client information into under the form of data units so called “packets" [1]. Yet, the fundamental specificity of packet networks is actually not directly related to the information form but to the transmission mode associated with packets. Indeed packets are sent without any preliminary reservation of resources between the source and the destination: at each hop their are transmitted according to the medium availability toward the next hop. Hence, the network bandwidth capacity is consumed on the fly according to the effective client packets to transmit.

This implies in particular that several packet transmissions can share the same link or path so far that they do not use them all the time (on contrary to circuits transmissions that should premiminary reserve). If the requirement of the clients sharing resources (link, path) are known, the sizing of this latter can be determinated statistically in order to serve adequately all clients. This kind of resource sharing, called statistical Time-Division Multiplexing (sTDM) is widely used by operators in the sense of an huge oversizing of transmission layers

Obviously the lake of reservation has some consequences if the effective client traffic can not be predicted. It introduces in particular some QoS issues. Basically in a first come first served network, when the transmission capacities of a packet link are excedeed, the

1. the generical term is Packet Data Unit (PDU) but those data units can be called frames, packets, cells or datagrams according to the context

concurrency involves delay if the packet is queued, and even loss if this latter is exceeded [2].

History In the 90s, with the growth of data taffics – first under the impulse of private companies and then with the Internet developpement – the packet paradigm has become prominent for operators. This led to a techno-economic battle between ATM [38] and IP [72]. Targeted to be the foundation for the “Broadband Integrated Service Digital Networks", ATM combines the connection-oriented networking approach with flexible traffic management functions to maximize bandwidth utilization while supporting a flexible set of services with different QoS guarantees [40]. Based on the TCP/IP protocol suite, the Internet provides a connectionless, best effort, end-to-end packet delivery service with the ubiquitous global connectivity. For several reasons, among which the finantial cost is probably the main, IP won this battle.

Then, to overcome the IP lake of QoS, Integrated Services (IntServ) [27] and then Differentiated Services (DiffServ) [63] were proposed [79] but almost no deployed.

During 2000s, IP core networks evolved toward IP/MPLS [75]

Paradigms Cohabitation – Next Generation PTNs

Packet Based Transport Networks (PTNs) [12] Unified Transport Network Solutions and CS/CO/PS cohabitation to get the benefits of the two worlds: hard QoS and carrier grade properties of the ciruit, and simplicity/flexibility of the packets.

MPLS-TP [22]. MPLS-TP started as Transport-MPLS at the ITU-T (see G.81XX series of ITU-T Recommendations), which was renamed to MPLS-TP based on the agreement that was reached between the ITU-T and the IETF to produce a converged set of standards for MPLS-TP.

Carrier Ethernet Ethernet is known for being used in local and campus computer networks, in both businesses and scientific institutions. Use the same technology in long-haul or carrier networks is presented to ease the integration of local/campus networks with carrier networks. Using the same technology also remove the requirement to encapsulate the customer’s Ethernet frames into another technology (which is necessary in the case of SDH/SONET or ATM), which would make the carrier’s equipment simpler. Because of the highly complex functionality offered by SDH/SONET and ATM technologies, the interface for these technologies is expensive. It is foreseen that employing the much simpler Ethernet

2. Queue scheduling mechanisms limit loss but they involve more complexity and transmission delays.

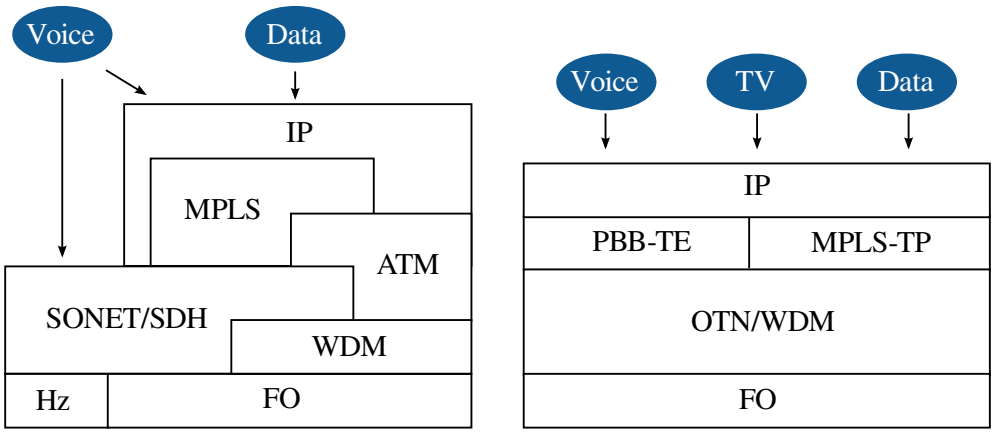
technology may reduce the cost of equipment and, as a consequence, the price of services. As far as customer services are concerned, the MEF has specifed Carrier Class Ethernet services [16], which gives to providers and their customers a reference for defining the terms of their contracts as well as allowing the certification of services and networking devices as MEF compliant.

Developments in recent years have added several new features which make Ethernet more suitable in carrier networks [3]. If the developments of interfaces with 10, 40 and 100 Gbps capacity constitute the main “bricks" forming the Carrier Ethernet, the significant Ethernet evolutions from local/private usage to metropolitan/carrier one are:

- In 2002, Provider Bridges (PB) (802.1ad, [42]). This evolution is in fact an extension of the Virtual LANs specification (802.1Q, [46]) that allows to insert multiple Virtual Local Area Nework (VLAN) headers into a single frame. The technique, also called “Q-in-Q" or “VLAN stacking", allows providers to isolate traffics by associating their own tags to services flows. PB allows to identify services up to 4094 instances.
- In 2007, Provider Backbone Bridge (PBB) (802.1ah, [44]). This evolution, also called “MAC-in-MAC", aims to improve the evolutionary potential of “Q-in-Q" by supporting several million of service instances
- In 2008, Provider Backbone Bridge Traffic Engineering (PBB-TE) (802.1ag, [4, 15, 43]). This evolution defines a connection-oriented version of Ethernet to makes it deterministic. It both removes Spanning Tree Protocol (STP), flooding and learning control mechanisms to allow Traffic Engineering. This way, a Carrier Ethernet network is operated in the same way than legacy transport technologies with a centralized controller. PBB-TE also adds OAM features to Ethernet.

If we consider the recent evolution of Transport Networks summarized in Fig 2.7, it is noticable that: IP is still both the unavoidable client layer of transport networks and the unavoinable server layer of applicative services ; the FO deployment is generalized ; SDH/SONET and WDM layers are naturally evolving towards OTN/WDM; finally two concurent proposal are in progress: Carrier Ethernet with the PBB-TE technology supported by the MEF, and MPLS-TP supported jointly by ITU and IETF.

3. Carrier Ethernet has especially forseen for Metropolitan Area Networks



(a) In the 2000s (b) Nowadays (2010-2015)

Figure 2.7 Transport networks evolution of general layouts

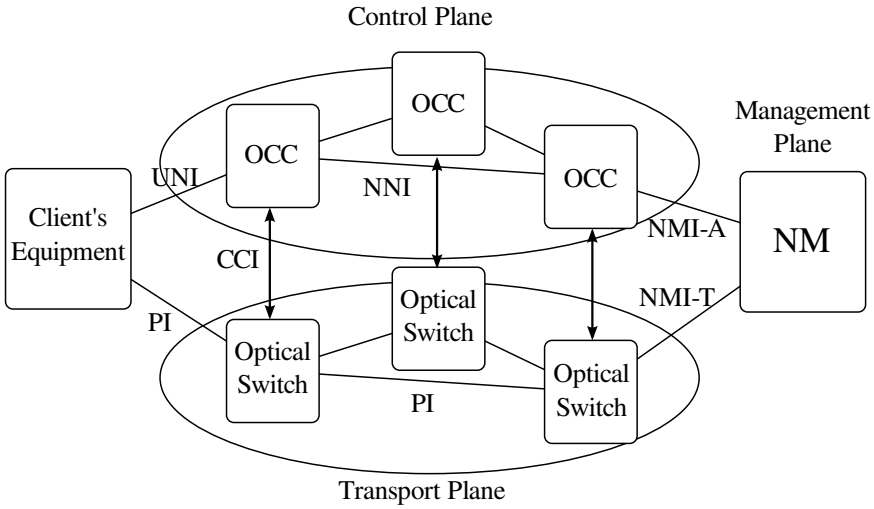


Figure 2.8 The ASON/ASTN distributed control plane
(Source: Wikipedia [78])

2.3 Transport Networks: Control & Management

2.3.1 Automatically Switched Transport Networks (ASTNs)

Towards a normalized Unified Control Plane (UCP) Since their emergence, legacy transport networks like SONET/SDH have been operated by using centralized Network Management System (NMS)[4] interacting with Network Elements (NEs) embeded in network devices with control protocol. In practive the important characteristic of this approach

4. also called Element Management System (EMS)

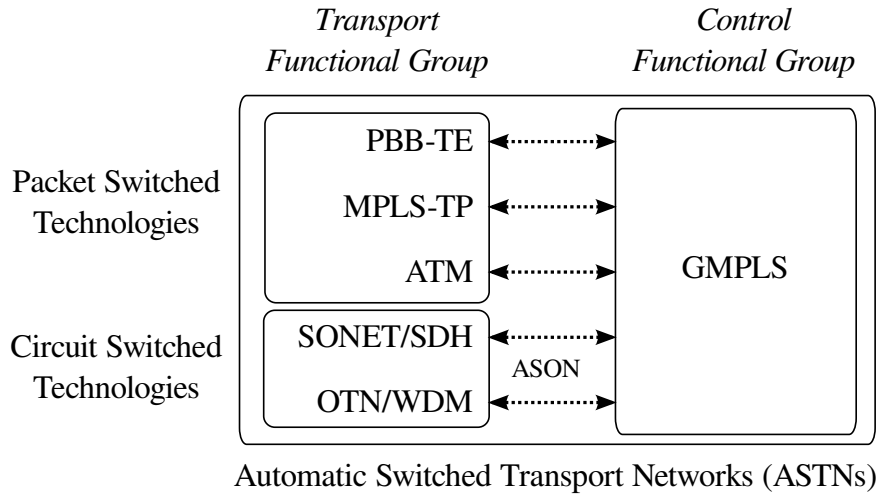


Figure 2.9 GMPLS is proposed as an Unified Control Plane (UCP) for Transport Networks

is to follow the vendors specifications both in terms of functionalities and interactions (interfaces and protocols) between NMS and NE controllers.

Considering the heaviness, complexity and sclerosing effects of the “NMS approach", many effort of normalization have been done for the build of *automated transport networks*, under the name of Automatically Switched Optical Network (ASON) and then Automatically Switched Transport Network (ASTN). The ITU-T describes the architecure in [14]. The distributed control plane is illustrated in Fig .2.8.

Generalized Multi-protocol Label Switching (GMPLS) Since 2004, GMPLS [59] is the distributed control plane envisioned for ASTN/ASON. This latter consistes in a collection of protocols and mechanisms adapted from MPLS to optical networks in order to automate them without “reiventing the wheel" . Ultimately, GMPLS promizes to operate all transport networks layers through a single normalized and Unified Control Plane (UCP) as shown in Fig. 2.9. This UCP is presented to come with significant benefits including management simplification, cost-efficiency, flexibility and error minimization.

GMPLS is in charge of: network topology discovery and resources discovery ; signaling, routing, address assignment ; connection set-up/tear-down ; connection protection/restoration ; traffic engineering ; Wavelength assignment. The GMPLS acronym designates the following protocols and mechanims (derived from MPLS standards):

- Generalized RSVP Extensions for Traffic Engineering (RSVP-TE) for signaling [20]
- Open Shortest Path First (OSPF) or Intermediate System to Intermediate System (IS-IS) with Traffic Engineering extensions for intra-area routing [51, 58]

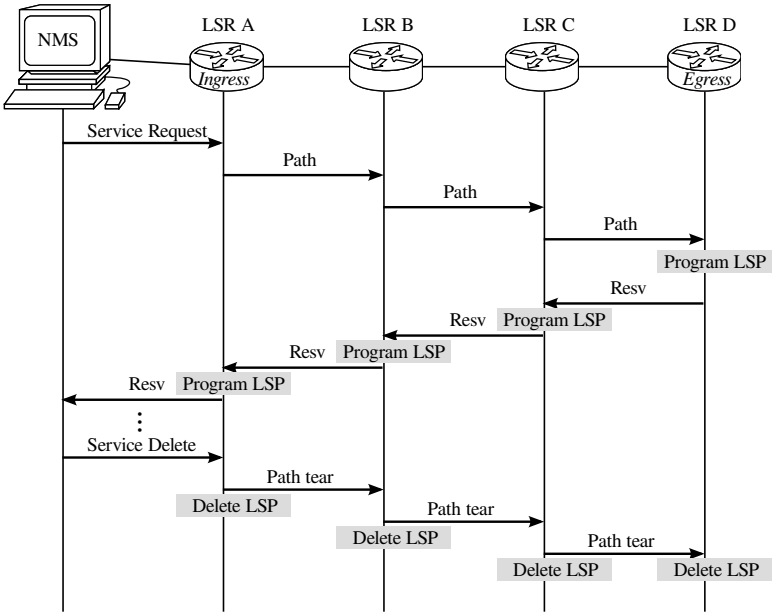


Figure 2.10 GMPLS messages exchanges for LSP establishment and teardown

– Link Management Protocol (LMP) or LMP-WDM for assorted link management and recovery functions [36, 53]
– Fast ReRoute (FRR) mechanism that establish local backup paths for the repair of LSP tunnels [67]
– GMPLS End-to-End Label Switch Path (LSP) Recovery [54]

LSP Establishment In GMPLS, a LSP establishment is initiated by the ingress Label Switch Router (LSR) sending an LSP *Setup* message to the next hop in the path of the LSP. The latter is determinated by looking at the explicit route of the LSP [5] or by computing the next hop toward the destination. The LSP is not created until it has been accepted by the downstream LSR which sends an LSP *Accept* message to supply the label that must be used to identify the traffic and to confirm the resource reservation.

The mechanism is shown in the Fig. 2.10 through the Resource reSerVation Protocol (RSVP) messages exchanges. The LSP Setup is forwarded downstream hop by hop until it reaches the egress. At each LSR the traffic parameters are checked to make sure that the LSP can be supported, and the next hop is determined. When the LSP Setup reaches the egress, it is converted to an LSP Accept that is returned, hop by hop, to the ingress. When the LSP Accept is received by the ingress and it has established its resources, it is ready to transmit data.

5. determinated by the administrator or by an external Path Computation Element (PCE)

LSP Maintenance Once an LSP has been established, we want to remain it in place until the service in no longer required. At the same time, if there is a failure within the network, we want the control plane to notice and recover the LSP if this latter is impacted. The original specification of RSVP is abominal for the way it handles these requirements as they related to packet flows. RSVP knows that the packet are forwarded according to the Shortest Path First routes derived by the IP routing protocols, so if there is a problem in the network it knows that the routing protocol will “heal" the network and find another route to the destination.

Therefore, RSVP specifies that the Path messages should be periodically retransmitted and that they should follow the same route toward the destination as used for the data. In this way, if there is a change in the network, the Path messages will adapt to these changes, new Resv messages will be returned, and resources will be reserved on the new routes.

However this phenomenon leaves the problem of resources reserved on the old routes – these resources need to be released so that they are available for other traffic flows. An explicit Release message could be used, but there may be no connectivity (the network may have become broken), and such an explicit release runs the risk of getting confused with reservation that are still required on part of the path where the old and the new routes are coincident.

To get around this problem, RSVP notes that if the upstream router is retransmitting Path messages, a downstream router can assume that the resources are no longer required if it does not see a retransmimtted Path message within a resonable period of time (usually set to three times the retransmission interval to allow for some loss of messages and some jittering of the retransmission timer). Moreover, RSVP specifies that the Resv message should be retransmistted so that it can detect faults and changes in the same way.

2.3.2 Recent Control Proposals

Path Computation Element Architectures (PCE)

Assuming the (G)MPLS control plane presented before, the initial idea of PCE architecture is to notice that path computation is a complex and greedy task that should be operated by a specially dedicated entity (a component, an application or a network node). Hence, path computation is no longer performed in the ingress node as in GMPLS. The component, so called the Path Computation Element (PCE) is able to master the establishment and maintenance of LSPs in a more generic manner.

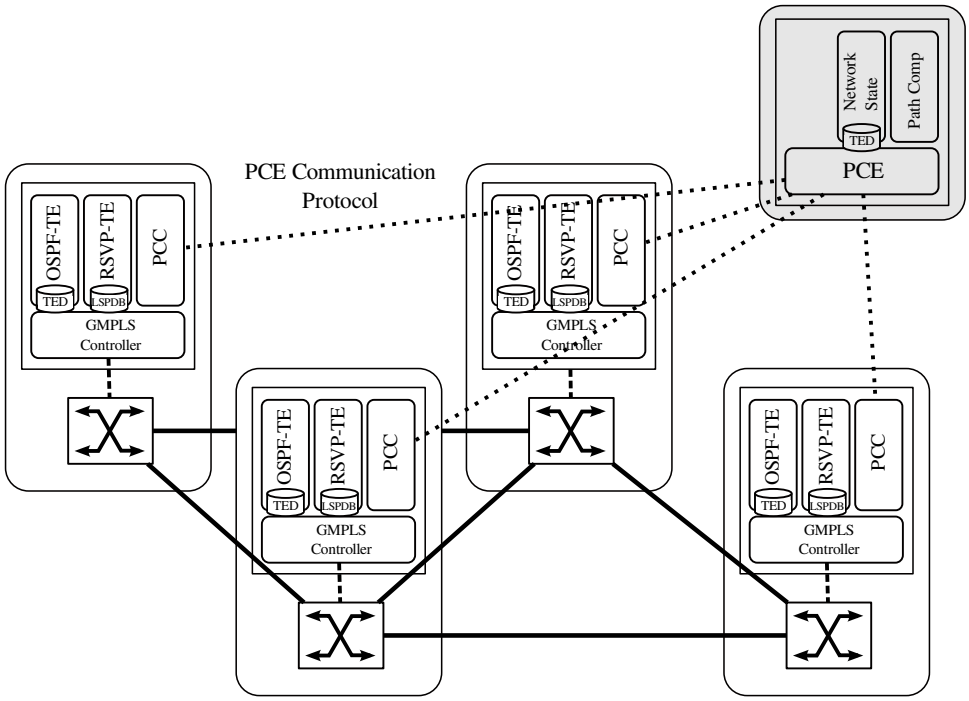


Figure 2.11 Example of GPMLS-controlled optical network with stateless PCE (Source: [62])

Then PCE architectures designates networks design solutions applying this decoupling between path computation and the rest of the control plane functions. To have an overview, significant surveys of PCE architectures have been done in [62, 68] where the various research trends of PCE architectures have been classified into general categories, namely: (i) single domain ; (ii) multi-domain ; (iii) multi-layer ; (iv) multi-carrier. Yet, the general principle applied in all PCE architectures if compared to (G)MPLS is clearly the centralization of the routing decision making which allows at the same time to centralize the complexity is a single point [6] and to determine almost optimal configurations (“all seing eye" design).

If GMPLS promises to automate Transport Networks, PCE promizes on its side to master the complexity of this automation by following a hierarchical organization. Globally, PCE over GMPLS proposal is the natural continuation of the bellhead paradigm toward automated CS networks. However during the last decade, other proposals – emerging outside of this paradigm – has been proposed to be applied to Transport Networks, challenging this “traditional way to evolve". We report two main proposals: *Autonomic Networking* and *Software Defined Networking (SDN)*.

6. or a small number of point in the case of a hierarchical arrangement

Autonomic Networking

In 2001, IBM producted a manifesto observing that the main obstacle to further progress in the IT industry is a looming software complexity crisis [52]. To answer to this problem this latter proposed a set of *autonomic computing principles* defining some *Self-Managing Systems* also known as *Autonomic Systems*.

An Autonomic System exhibits some self-management properties called self-* properties. As noticed in [31], the four properties exposed by IBM in 2001, constitute the most widely recognized elements of Autonomic Systems within autonomic computing research area, namely:

1. *Self-configuring* This property refers to the capacity of the system to configure and reconfigure itself in accordance with high-level policies in a changing environment. It involves the ability of both the new component and the system to instal, configure and integrate when a new component is introduced to the system. The component would be able to incorporate itself seamlessly, and, the system to adapt itself to its presence. The end system could then make use of this component normally or modify its behavior accordingly
2. *Self-optimizing* The objective of self-optimization is to enable efficient operation of the system even in unpredictable environments. An autonomic computing system will proactively seek opportunities to make itself more efficient in performance and cost. For this, the system should be aware of its ideal performance, measure its current performance against the ideal and have strategies for attempting improvements
3. *Self-healing* This property consists in the capacity of discovering and repairing potential problems to ensure that the system runs smoothly. It may be achieved by predicting problems and taking proactive actions either to prevent failures or to reduce their impact
4. *Self-protecting* Self-protection designates the ability of the system to protect itself from what compromises it from achieving its goals. It involves the protection from malicious attacks, intrusion tentative or inadvertent failures

To reach those self-* properties, an autonomic system considers managed elements and autonomic controlers, where the adaptation process follows an Autonomic Control Loop (ACL) – also called Monitor Analyse Plan Execute Knowledge (MAPE-K) loops. The classic IBM model of the ACL is depicted in Fig. 2.12.

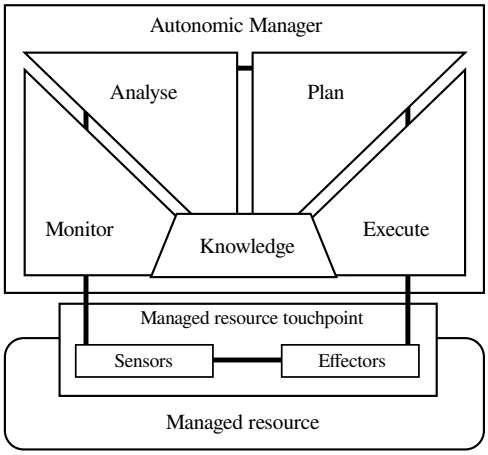


Figure 2.12 IBM's reference MAPE-K loop (Source: [52])

Moreover, an autonomic system is defined by the notion of High Level Objectives (HLOs). These latters are some externally defined global goals expected to be reached by the system through the ACL rounds.

The distinction between autonomic communication and autonomic computing is provided in [30]:

- *Autonomic computing* is seen as a way of reducing the total cost of ownership of complex IT systems by allowing reconfiguration and optimization to proceed on an ongoing basis driven by feedback on the system's ongoing behavior. It combines a technological vision with a business rationale for increasing the coupling between business goals and IT services.
- *Autonomic communication*, by contrast, generally refers to all these research thrusts involved in a deep foundational rethinking of communication, networking, and distributed computing paradigms to face the increasing complexities and dynamics of modern network scenarios. The ultimate vision of autonomic communication research is that of a networked world in which networks and associated devices and services will be able to work in a totally unsupervised manner, able to self-configure, self-monitor, self-adapt, and self-heal—the socalled self-* properties. On the one hand, this will deliver networks capable of adapting their behaviors dynamically to meet the changing specific needs of individual users; on the other, it will dramatically decrease the complexity and associated costs currently involved in the effective and reliable deployment of networks and communication services.

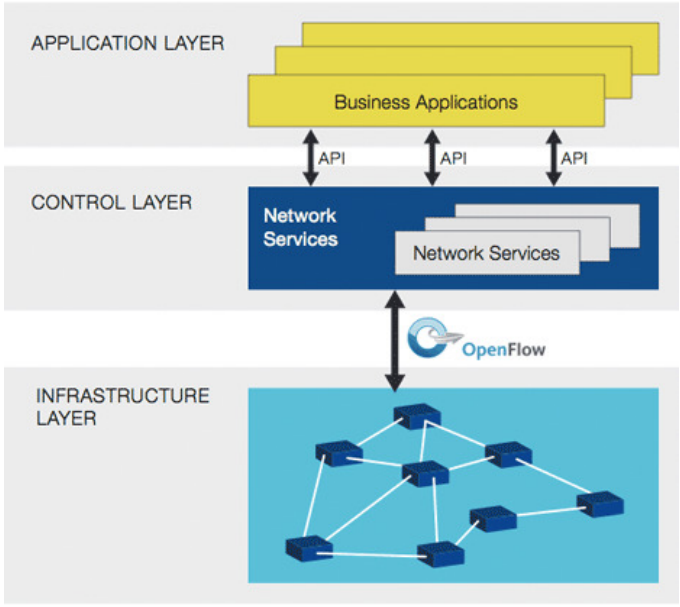


Figure 2.13 The SDN system architecture (Source: [37])

Between 2001 and 2010, many proposals emerge in autonomic networking under the form of architectures and frameworks. [60] surveys and classifies the mains and allows to get comprehensive view of the research field.

Software Define Networking (SDN)

We exposed previously in the section that PCE is a hierarchical control approach overhanging GMPLS protocols. Then, we have seen that Autonomic Networking constitutes a research field which the main purpose was to master network complexity under a specific automation paradigm. Additionally, Software Defined Networking (SDN), initially proposed for experimental networks and then Data Centers, is curently proposed as a Transport Networks solution. In this Use Case, SDN can be presented as follows regarding to the previous proposals: (i) its global purpose is to master the network complexity, like Autonomic Networking ; (ii) it follows a hierarchical control approach, like PCE ; (iii) yet it does not assume GMPLS, hence it concretely calls into question the PCE/GMPLS proposal.

According to the Open Networking Foundation (ONF) [37]: “SDN is an emerging architecture that is dynamic, manageable, cost-effective, and adaptable, making it ideal for the high-bandwidth, dynamic nature of today’s applications. This architecture decouples the network control and forwarding functions enabling the network control to become directly programmable and the underlying infrastructure to be abstracted for applications and net-

work services – see Fig. 2.13. The OpenFlow protocol is a foundational element for building SDN solutions."

The SDN architecture exhibits the following properties:
Directly programmable – Network control is directly programmable because it is decoupled from forwarding functions.
Agile – Abstracting control from forwarding lets administrators dynamically adjust network-wide traffic flow to meet changing needs.
Centrally managed – Network intelligence is (logically) centralized in software-based SDN controllers that maintain a global view of the network, which appears to applications and policy engines as a single, logical switch.
Programmatically configured SDN lets network managers configure, manage, secure, and optimize network resources very quickly via dynamic, automated SDN programs, which they can write themselves because the programs do not depend on proprietary software.
Open standards-based and vendor-neutral – When implemented through open standards, SDN simplifies network design and operation because instructions are provided by SDN controllers instead of multiple, vendor-specific devices and protocols.

Applied to the Transport Networks automation, SDN is primarily a centralized control approach to be opposed to the GMPLS distributed control plane. Hence, the question is: what makes SDN a fundamentaly different proposal than traditional NMS centralized approaches ? If compared roughly to an NMS architecture, SDN basically replaces embed Network Elements (NEs) with standardized northbound interfaces receiving controller instructions with the openflow protocol rather than another control protocol. On this basis, the network can be said programable since the controler is itself programmable (not a revolution from this angle).

A significant survey effort concerning SDN has been done in [65]. Beyond the technical characteristics and the potentialities provided by the abstraction of network layers, we consider that the fundamental SDN underlying point is the *standard-based vendor-neutral* approach of control, opposed to the vendors specific NMS proposals. Indeed, the question of the control plane opening is fundamental in the Transport Network context because of the profound implications onto the principal Internet actors: Vendors, Network Operators and Content Providers. The question of the control standardisation is in fact the main reason of the huge amount of R&D activities related to SDN.

In a Transport Network context, SDN becomes in the center stage for the same critical reasons than GMPLS was few years ago [29, 33], and it raises the same questions than this latter in terms of final deployements. The SDN rising phenomenon in Transport Networks research activities is the tangible manifestation of the economic battle between Network Operators and Content Providers for the control of the infrastructure.

2.4 Conclusions

Ongoing evolution of PTNs

To cope with the mutiplicity of services requirements, operators Transport Networks evolved considerably since the 80's. The decisive introduction of Packet Switching with IP has firstly seemed to face all needs very simply through the overdimensioning of networks transmission capacities. However, since the IP transition, mechanisms from the legacy telcos paradigm have been re-introduced into PTNs: (i) the connection oriented transportation – for determinism ; (ii) introduction of bandwidth reservation, it is to say the reconstitution of circuits upon packet switching – for traffic engineering abilities and hard QoS transportation ; (iii) legacy circuit protection mechanisms – for resiliency to failures and network availability guarantees.

Two major proposals embody this trend: (G)MPLS architectures, in particulars those based on its Transport Profile specifications [22, 32, 77] (i.e. MPLS-TP), and Carrier Ethernet [26] (i.e. PBB-TE). This trend is actually nothing else than a return to the ATM purpose, namely integrate Packet and Circuit Switching in an unified architecture providing all types of services, including a distributed control plane and the OAM features required by operators.

For now, the IP/MPLS and GMPLS proposals are the most mature ones through the suite of standards developed for control, signaling and OAM features. Yet to meet the carrier grade requirements, we have seen that several protocols and mechanisms must be added to MPLS. Finally, the implementation of an “all automated" (G)MPLS solution involves many components and interworking mechanisms and requires complex (thus exensive and error prone) statefull nodes. The addition of a separated PCE entity for the path determination is still increasing the heaviness of (G)MPLS architectures with always more states, mechanisms and protocols.

Ongoing evolution of PTNs' control plane

It is interesting to notice that until now, transport networks' control planes always relied on the proactive routing paradigm for the determination of paths from end-to-end [7]: (i) it is obvious in IP networks using widespread protocols such as OSPF or IS-IS ; (ii) yet the same applies to MPLS/GMPLS that rely on similar protocols to fed their Traffic Engineering (TE) routing tables used to ultimately establish LSPs [8] ; (iii) as well, to provide a scalable Ethernet control plane the proposals assumed LS routing – see Transparent Interconnection of Lots of Links (TRILL) [70, 71] and Shortest Path Bridging (SPB) [34, 45] ; (iv) in its time ATM proposed its LS routing specification under the name "PNNI routing". Technically, the main reason of the systematic choice of proactive routing is the operators' *imperative of scalability*.

These days, it seems to be always assumed that some combination or mix of Ethernet (PBB-TE) and/or MPLS-TP and maybe GMPLS [59] is the definitive – but highly complex – answer to creating that optimum highly integrated "Next Generation Network" architecture that can be used to provide any service any customer might require. Maybe, it is worth considering a complementary approach focusing on simplicity – as pointed out since several years by lucid experts [39].

To cope with the network complexity of configuration and maintenance, numerous works aim to build "self-managed networks" by applying the Autonomic Computing concepts defined in [52] to networking. This should allow to adapt the network configuration to the context evolution according to some predefined High Level Objectives (HLOs). For operators, the main interest of Autonomic Networking is the minimization of human interventions in so far they are both expensive and error prone. However, no autonomic proposal led to an implementation in real operators networks so far, with the notable exception of the 3GPP SONs mechanisms in Radio Access networks [18, 69]. Until now there is no significant commercial proposal for a generic, extensible autonomic transport network architecture. Among many reasons of the low interest in core segments, it possible to bring out the relatively low amount of effective "objects" to be managed for the moment [9].

7. namely Link-State (LS) or Distance-Vector (DV) protocols and their associated algorithms
8. on the base of extended Interior Gateway Protocols (IGPs) specifications such ISIS Extensions for Traffic Engineering (ISIS-TE)) [58]
9. understand low amount of transport entities due to aggregation

Finally, Software Defined Networking (SDN) constitutes at this time the most promizing proposal toward an unified, multi-vendors and multi-domain automation of metro/core transport networks. Not so such for the novelty of its goal than for the heavy actors supporting its (Google among others).

LOCARN: an alternative to focus on simplicity

In 2010, by observing the overall complexity of the control structures involved in (G)MPLS architectures and the absence of an autonomic practical deployement among core networks, an innovative packet transport solution named LOCARN (i.e. Low Opex and Capex Architecture for Resilient Networks) has been proposed. LOCARN aims to bring simplicity while maintaining the functionalities and performances of current transport networks. LOCARN is both very simple by components design and very simple to operate. It relies on the interworking of three basic mechanisms which require low computation complexity and almost no states in nodes. A fundamental difference of LOCARN with all the presented solutions is to not rely on any Link-State (LS) or Distance-Vector (DV) routing protocol and algorithm for the determination of paths. In LOCARN, the paths determination process: (i) is specific to a client service (defined at least as a peer of access points) ; (ii) does not need to maintain a global view of the routing domain – paths are not computed but simply discovered, characterized and compared. Those radical differences leads LOCARN to bring very beneficial properties and behaviors if compared to legacy routing proposal in Transport Networks.

The next chapter exposes the architecture in more technical details before to discuss its characteristics and finally position it in the technological neighborhood.

Chapter 3

The LOCARN Architecture

3.1 Technical Overview

Terminology Since the LOCARN solution is constituted by an assembly of interworking mechanisms involving the data plane, the control plane and OAM features, we refer to it as a “Network Architecture".

3.1.1 How LOCARN works

LOCARN is a flat packet network architecture providing point-to-point bidirectional communications through the definition of its own data plane and control plane mechanisms, in a complete agnostic way with regards to the IP stack. The functional architecture, illustrated in Fig.3.1, is composed by two kinds of functional nodes: (i) *Edges Nodes (ENs)* that constitute ingress/egress nodes of a LOCARN transport domain (ii) *Transit Nodes (TNs)* that solely operate packet forwarding. The functional architecture also defines: (i) *Access Ports (APs)* that are external ENs ports – they can be physical or logical ports according to the implementation guidance (ii) *Services* that are end-to-end bidirectional channels [1]. Services provide customer data with the transport of its client information between two APs across the domain. They are composed by two *Reference Points (RPs)* sharing a common service identifier: a *Connection Origin Reference Point (CORP)* and a *CORP*. The only management operations required in LOCARN is registration/unregistration of RPs using

1. the LOCARN architecture forms a layer network in the sense of the ITU-T G.805 recommendation: “A topological component that represents the complete set of access groups of the same type which may be associated for the purpose of transferring information"; in the ITU transport terminology, a LOCARN service is a point-to-point *transport entity*, i.e.: “an architectural component which transfers information between its inputs and outputs within a layer network"

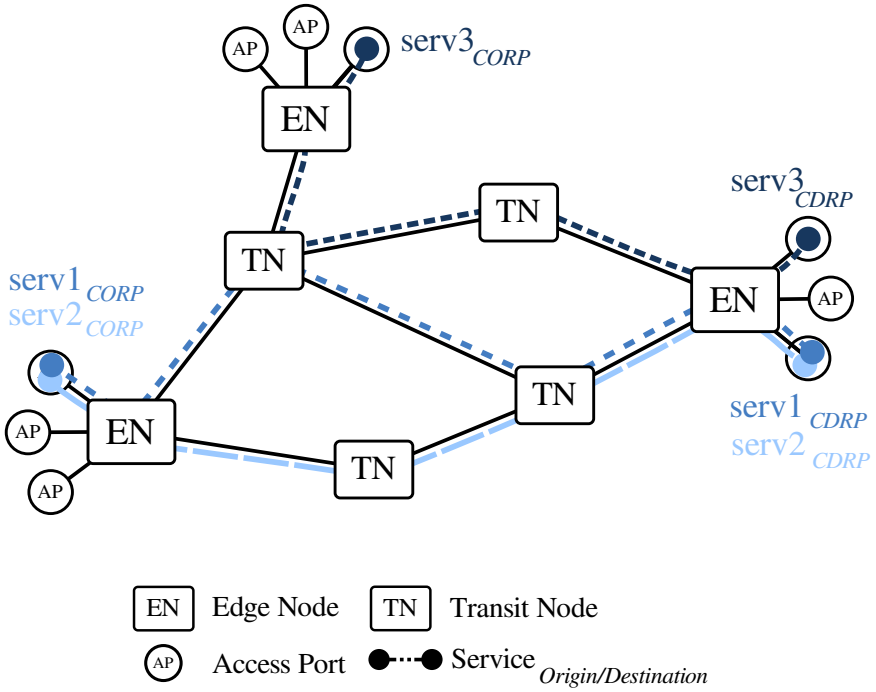


Figure 3.1 The LOCARN architecture illustration

unequivocal services names among the domain. Then LOCARN is in charge of the establishment, maintenance and adaptation of declared services over time.

The LOCARN architecture is composed of three main mechanisms:

Autoforwarding Autoforwarding designates a packet relaying mechanism that uses exclusively information contained in the packet itself for the determination of the next hop interface. In LOCARN, the data packets are autoforwarded from end-to-end by carrying in headers the complete sequence of output ports to cross from the origin to the destination (including the AP as the last port). Autoforwarding leads to a drastic simplification of the data plane technical requirements, both in terms of computation and memory space. Indeed, the autoforwarding header is just added at data packets encapsulation in the ingress EN whereas successive TNs forward packets by obtaining the output port directly from this header. A sequence cursor is incremented at each step of the course to indicate the header position to read. An autoforwarding data plane does not require to store any Forwarding Information Base (FIB) in TNs neither a fortiori to perform any table lookup operation.

Enhanced flooding To obtain the autoforwarding header information included at the data packets encapsulation, an EN runs a source routing process for each CORP registration. A

LOCARN source routing process can be decomposed into six steps: (i) a service path request is flooded across the network [2] – a Time To Live (TTL) mechanism allows to bound the packets propagation ; (ii) over time, some request packets – arriving from distinct paths – reach the EN where the corresponding Connection Destination Reference Point (CDRP) is registered ; (iii) for each request packet, the “destination EN" returns an answer packet to the “origin EN" [3]. Each answer message carries to the origin the autoforwarding information of a specific path EN (to do so, request packets have to memorize the crossed output ports along their course); (iv) along the backward course, answer packets take the opportunity to collect statistics about the crossed interfaces (bandwidth usage, queuing usage...); (v) then the origin EN typically receives numerous answer packets where each one provides an autoforwarding port sequence associated with end-to-end statistics about the path. Let us notice that cyclic paths discovery is prevented during the flooding propagation by deleting packets that followed a loop ; (vi) finally the EN origin selects the best path by successive comparisons between the arriving path and the one that is currently used. In LOCARN, source routing is thus purely achieved through paths *discovery* and *selection*. No routing algorithm is involved, no routing computation or routing tables either, and no convergence times: a path has changed from the moment that the encapsulation function has been modified.

The decisive specificity of LOCARN routing lies on the fact that: (i) a routing process is associated to each service – “serv1" and “serv2" illustrated this in Fig.3.1 by choosing distinct paths ; (ii) among the (large) amount of discovered paths, a service will typically favors the less used paths. To do so, the service’s path selection is partially or totally based on the discovered paths’ meta-information collected (i.e. interfaces statistics about bandwidth consumption and queuing loads) which actually reflects the level of effective resources usage along the path [4]; (iii) for each service, the source routing steps are periodically relaunched to adapt the active path over time to any kind of noticeable evolution between the endpoints (referred as “path-optimizations"). Thus, on one hand each service self-adapts periodically according to its perception of the network state between its endpoints; and on the other hand a service adaptation is itself noticeable by other services within the domain, and may trigger

2. each LOCARN nodes, ENs and TNs, contributes to the discovery by duplicating the request packets through all nodes interfaces except the incomming one

3. each answer packet is autoforwarded along the reverse corouted path from destination to origin by using the request packet memorization of the crossed input ports along the request course

4. several routing policies can be implemented. The use of the available bandwidh criteria tends to balance loads for example whereas the end-to-end delay criteria can be used to provide adapted path for clients traffics sensitive to delay (on its side, a path end-to-end delay is obtained by noticing the Round Trip Time (RTT) at the path answer arrival time).

their own adaptation [5]. Finally, over successive path-optimizations, LOCARN tends hollistically toward a global, opportunistic and very adaptative distribution of client traffics among the domain without involving any complexity in the achitectural design. In-depth studies regarding LOCARN dynamic behaviors are not the focus of this thesis (both positive behaviors like overall path arrangements and negative aspects like possible erratic behaviors).

End-to-end fault detection The periodical relaunch of source routing steps exposed above make services able to recover if the connectivity has been lost for any reason. Yet, the optimization periodicity is envisioned at least around several seconds making the recovery time relatively long. With regards to the operators Transport Networks requirements in terms of resiliency, each LOCARN active path is continously checked by very frequent round trips of specific small OAM packets (i.e. Operation And Maintenance); the lack of three OAM answers at the CORP means that a fault has occured. These packets exchanges, initiated by CORPs, are strictly transfered like data packets with autoforwarding. The interval of OAM sends determines the detection time, it must be envisionned around several milliseconds to reach carrier grade networks.

3.1.2 General Pros and Cons

LOCARN's Benefits: Properties and Behaviors

A very simple design and mehanisms LOCARN proposes simple nodes design based on minimal computation and storage: low states in edge nodes and no state at all in transit nodes. Hence the architecture is easy to implement both into software or hardware components with low technical requirements.

Tunable resiliency to failures LOCARN copes with resources failures, ensuring the continuity of client information transit with the minimum of impact. The LOCARN resiliency relies on a purely reactive restoration process without any preliminary configuration of any protection path. On a service fault detection (related to the settings of OAM exchanges), the CORP simply re-launch the source routing processe in order to restore the path. This reactive approach is possible because the path discovery is very quick (a path between an origin and a destination is basically obtain during its RTT). Doing so, LOCARN avoids the waste of network transmission capacity if compared to a dedicated protection path. More-

5. a service adaptation leads to a more or less significant redirection of the client traffic which impacts interfaces statistics that will be collected by future services' discoveries

over LOCARN is intrinsically able to cope with multiple failure uses cases: if at least one path exists between the client's service endpoints, it can be used for recovery.

Continuous adaptation to both client and server layers LOCARN has been conceived to handle both dynamic clients demands and unstable/evolutive infrastructures. Indeed, when the client demands change, the services end-to-end paths adapt through the domain according to the evolution of effective infrastructure evolution of use. We understand by dynamic client demands the variability of traffics loads as well as the variability of the traffix matrix itself (irregular arrival/departure of client communication). We understand by evolution of the infrastructure the addition of links or nodes (changing the infrastructure topology) or the increase of links bandwidths as well as on the decrease of the network transporation resources. Finally, LOCARN constitutes a network layer in continuous adaptation of end-to-end paths in order to continually adjust the traffics where are the available resources.

About LOCARN adaptiveness and settings:

- Globally, the *sensitivity* of the LOCARN routing adaptations depends on: (i) the services path selection criteria. For example the routing objectives of a service can be to optimize the *end-to-end delay*, the *jitter*, the *bandwidth usage* [6], or to define rules combining of those criteria ; (ii) the setting of the interfaces' statistic information collect (frequency of measurements and the frames duration for the statistic aggregation of the information).
  It is suitable that communications can establish or released without the involvement of adaptation processes (from other services present witin the domain) if their impact on the infrastructure is lows, typically quick or light communications (a threshold must be found). On the contrary, a service which needs a high datarate punctually must involve some paths shifts among the domain. On its side, the *reactivity* of the adaptiveness is tunable with the Optimization Interval (OI) – Interval of flooding relaunch for a same service.
- The routing *level of opportunism* depends on the *sensitivity* and *reactivity* allowed by its settings, but not only. To be as opportunistic as possible, LOCARN depends on the *elasticity* of the source routing (i.e. the ability to find longer paths than the shortest one). The elasticity can be tuned with the TTL setting.
  It must be noted that the opportunistic aspect of LOCARN is solely benefitial in

6. to follow a such objective, the tuning of the end-to-end "occupancy score" threshold is decisive

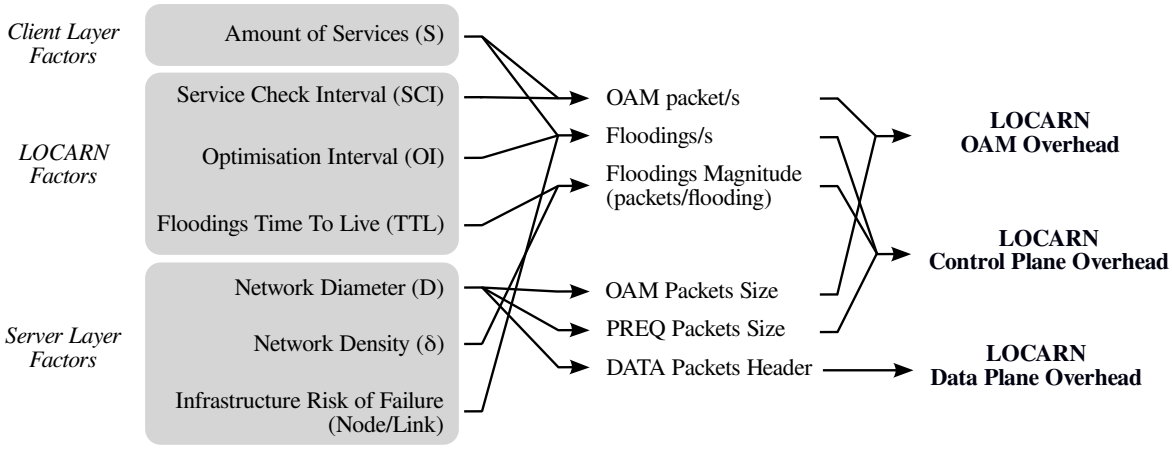


Figure 3.2 Factors of LOCARN overheads

meshed topologies. Further, the more a network is dense, the more the discovered paths are numerous, and thus the more LOCARN routing is opportunistic [7]. However, as exposed in chapter 5, the flooding's overhead increase with the network density as the counterparty to the opportunism.

Plug&Play guidance The addition of new services simply consist in the declaration of the Reference Points with appropriate indentifiers. Once it is established, each service is included in the network domain adaptation process without requiring any planning-configuring step. The LOCARN operational simplicity is known as "plug-and-play", because the network is ready to work as soon as the devices are plugged onto the underlying layer whereas a service is ready to transmit immediately from the moment its endpoints have been declared.

Minimal financial cost The simplicity is besides a mean to involve low financial costs: the nodes low technical prerequisites mean low CAPital EXpenditure (CAPEX). On the other hand, operational simplicity and adaptiveness lead to minimize the OPerational EXpenditure (OPEX). Finally, the data plane simplicity could allow an important energetic cost redution (i.e. OPEX) since both FIBs and look-up operations are eliminated from the end-to-end transfer of data packets.

LOCARN Issues: Overheads and Scalability

As stated before, LOCARN is simple, stateless, dynamic and resilient. Yet those properties do not come without any counterparty. The couterparty is the overhead involved by the achitecture with the significant production of non client packets over time. This packet production must be estimated to evaluate the overhead bandwidth consumption and enssure that it does lead the architecture to be inneficient. A big difference of LOCARN with other protocols is that the involved overhead is not solely related to network dimensions but it also depends on the number of communications (services) declared within a domain. We distinguish three kinds of overheads (see Fig. 3.2) according to the three LOCARN mechanisms presented above:

Data plane overhead (autoforwarding fields) Because of autoforwarding, the LOCARN data packets header's size depends on path lengths (each path hop requires an autoforwarding field). Therefore autoforwarding could a priori presents a scalability problem. Yet, in practice the LOCARN framing scheme actually involves smaller headers than common packet technologies. The explanation is simple: in LOCARN a data header solely contains the output ports from a source to a destination. It does not include any node identifier neither for source, destination or intermediate nodes. Hence, a 20 hops path [8] involves the storage of 20 ports in data headers. If the implementation assumes for example that the identifier of transit ports is encoded on 1 byte (meaning a maximum of 256 transit ports per Transit Node), then the header remains smaller than a MAC-in-MAC encapsulation of Carrier Ethernet frames (802.1ah) and it remains equivalent to five MPLS label stack entries [74]. Hence, data headers must not be neglected for an accurate calculation of the total overhead but it is not critical beside overheads related to control and OAM message generation.

Control plane overhead (flooding propagation) The flooding overhead is definitively a critical point to study in LOCARN. Let us first notice that it can not be envisioned to let the flooding process terminate by itself since the number of generated messages would exceed a reasonable number even for small networks. In LOCARN the floods propagation is simply limited by the mean of a TTL mechanism included to the flooding messages treatment. This approach raises the question of the TTL tuning. We easily understand that the TTL must be at least equal to the shortest path(s) length between an origin and a destination to allow the

7. LOCARN has almost no interest in a ring topology for example neither in terms of path adaptation and recovery

8. that can be considered as a very long path across a single transport domain

establishment of a connectivity; whereas each TTL increment increases the amount of path discovered whose the beneficial consequence is a higher potential for path optimization and for recovery. For simplification, we consider in following sections a global TTL applied to all services path discoveries. The selected value is comprised between the network diameter and a reasonable upper bound to be estimated.

OAM overhead (end-to-end exhanges for fault detection) The OAM exchanges between service's end-points is another critical factor of overhead cost in LOCARN due their high frequency. Indeed, to meet high resiliency level, OAM messages must be sent with high frequency – in practice, by setting an interval around 10ms or 15ms for sub-50ms service protection.

3.2 LOCARN Positionning

3.2.1 Two Conceptual Ancestors

The Dynamic Source Routing Protocol (DSR)

Commonalities The Dynamic Source Routing (DSR) protocol is an experimental protocol for ad-hoc wireless networks [48, 49] that is conceptually close to LOCARN by sharing the two following principles. First, *stateless nodes* are employed relay data packets from end-to-end with an autoforwarding mechanism – nodes rely on the information contained in packets themselves to relay them from end-to-end deterministically. Secondly, *reactive source routing* is used to *discover* the network topology *on demand* from a source node (specifically, DSR relies on 802.11 broadcasts).

Their approach is actually orthogonal to well widespreads approaches used in intradomains and based on proactive routing – Link-State (LS) and Distance-Vector (DV). Indeed, in proactive routing protocols, some packets are continuously disseminated within the routing domain in order to maintain a global knowledge of the network topology in each node. Then, on the basis of its information, each *statefull node* has to *compute* routes to other nodes of the domain by using a shortest path algorithm (like Bellman–Ford for DV or Dijkstra's for LS).

Differences If both LOCARN and DSR involve the same concepts, the differences of designs are due to different assumptions and final objectives, which lead them to exhibit very different technical properties.

The use case assumption – DSR is envisioned for wireless ad-hod networks to assume the resource scarsity of a such environement. The forseen interest of DSR is to avoid the continuous packet production of proactive routing protocols because of their energetic cost. This leads DSR to launch routes discoveries as least as possible, and even to maintain some “softstates" in nodes to this regard. On its side, LOCARN is envisioned for transport networks where resources are much more abundant and where the energetic issues are different (Fiber Optics instead of the radio medium). Hence, overhead due to floodings discoveries is less critical since links commonly exceed 10Gbps.

The IP stack assumption – DSR assumes to run over the IP stack which leads it to encapsulate IP addresses for the end-to-end data forwarding. On its side, LOCARN assumes to be totally agnostic to IP ; it solely store enpoints identifiers and intermediate ports in autoforwarding headers. It allows a very minimal data header size which is particularly suitable in transport networks where data headers have an important impact on the network overhead because of the datarates magnitudes.

The routing policy – In DSR, like in the proactive routing protocols cited above, the widely used routing policy consists in the choice of the shortest path. It is well established that saving resources is the main concern in wireless adhoc networks. On the contrary in LOCARN we propose an alternative routing approach for transport networks which can be described as “opportunistic to the QoS" and service oriented. Indeed, to favor the delay, jitter or bandwidth of a client service, LOCARN is able to use other paths than the shortest path, i.e. elastic paths. Moreover the routing decision making is done by each individual services; it means that distinct paths can be selected for services declared on the same nodes (in particular it allows load balancing).

The APLASIA Architecture

The second related work that is conceptually close to LOCARN is the APLASIA architecture [76]. APLASIA was inspired from the LOCARN initiative and is notably based on the same primary concepts, namely autoforwarding and routing based on paths discoveries by means of network floods. Yet, in the APLASIA design, those mechanisms are used in a different way, providing ultimately very different properties and behaviors than LOCARN. Among other differences, the Adaptive Probabilistic Flooding (APF) [23]) algorithm is employed to minimize the message generation involved by flooding processes (reducing at the same time the amount of discovered paths).

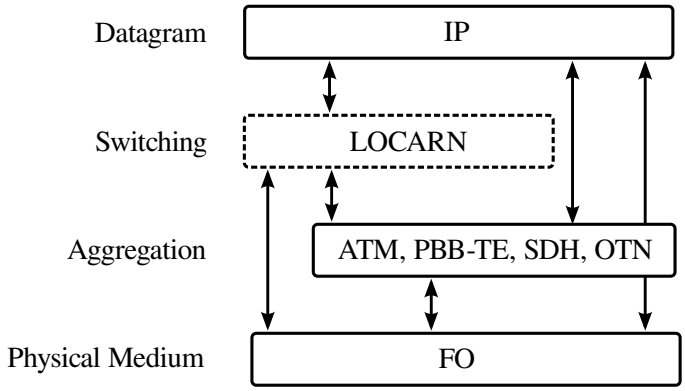


Figure 3.3 LOCARN best position as a packet transport architecture

3.2.2 LOCARN as a Packet Transport Architecture

Transport networks traditionally designate purely circuit TDM layers like SONET/SDH/OTN. In the current today's packet-oriented world, transport networks are more and more constituted of packet-based equipment. As a Packet Based Transport Network (PTN) solution, LOCARN is an alternative to several packet technologies: (i) "Carrier Ethernet and layer 2 routing solutions" that mainly forseen for metropolitan networks ; (ii) IP/MPLS architecture through its suite of standards for Traffic Engineering. The more suitable position for LOCARN as a PTN solution for operators is presented in Fig. 3.3.

Carrier Ethernet and Layer 2 Routing

For the design of packet Transport Networks, an idea was to make the Ethernet technology evolve towards "Carrier Ethernet". Behind this denomination, the goal is to make the Ethernet technologie able to scale for operators backbone usage. To do so, PBB-TE broadcasting and learning processes are suppressed [9] from the original bridged Ethernet mechanisms because of their scalability issues ; whereas on the other hand, in the forwarding plane, the ethernet header is enriched for VLANs [10] stacking under the name of "Q-in-Q" 802.1q [46], and for MAC-in-MAC encapsulation under the name of PBB 802.1ah [44].

Those evolutions definitively solve the Ethernet scalability issue, yet they also reduce Ethernet to a forwarding plane that does not establish communications by itself anymore. It is to say that it can/must be operated in the same way than legacy transport solutions

9. as well as in PBB
10. VLANs are Logicaly configured subparts of a network. The main interests of VLANs in transport networks are the simplication of packet switching (reduce delays) and traffics isolation.

with a NMSs approach or by adding a control/signaling plane like GMPLS. Some efforts attempted to make Ethernet both scalable and automated by introducing an IGP routing protocol (namely IS-IS) at the Ethernet level : TRILL from the IETF (Rbridges) [70, 71] and SPB from the IEEE (802.1aq) [34, 45].

IP, Ethernet and MPLS proposals for Transport Networks

In traditional intra-domain routing protocols such as OSPF [61] and IS-IS [66], which route traffic along the shortest paths, failures cause at least two problems: first, new link state information needs to propagate throughout the network and new paths need to be found. The transient state before routing stabilizes can last up to 30 seconds [21], [41], unacceptable for real time applications such as voice over IP. Second, some links need to carry rerouted traffic and can become congested. Studies suggest that as much as 80% of congestion in the backbone is caused by failures [47]. MPLS helps solve the first problem by quickly rerouting traffic from the primary path to a protection path [67, 75]. But it still cannot guarantee to prevent congestion. So, to reduce the chance of congestion, backbone networks are hugely over-provisioned.

The most widespread packet transport solution is the IP/MPLS technology. Both IP/MPLS and LOCARN aims to establish and maintain end-to-end connectivities in a transport networks. However, they achieve these purposes with rather different ways that lead them to exhibit different properties. Typically, MPLS and GMPLS transport architectures combine the following standards for control, signaling and OAM (see Table 3.1):

- an IGP routing protocol collects topological information: commonly ISIS-TE [58]
- a signaling protocol for the establishment of end-to-end communications, commonly RSVP-TE [20] or Label Distribution Protocol (LDP) [19]
- a protocol for the local detection of failures, commonly Bidirectional Forwarding Detection (BFD) [50]
- the FRR mechanism and signaling extensions for a local paths protection [67]

We see here that IP/MPLS architectures establish and maintain communications through many protocols and some quite complex mechanisms (involving computation and memory space). On the other hand, LOCARN establishes and maintains communications very simply by using few mechanisms to the cost of more significant overhead production that is finally the couterpart of the simplicity and flexibility. In fact, LOCARN trades the *bandwidth consumed* by its overhead for *simplicity*, *self-adaptiveness* and *resiliency*. We believe

	LOCARN	(G)MPLS standards
Topology Discovery	Path Discovery (Periodical)	Link-State diffusion ISIS-TE [58], OSPF-TE [51]
Connection Establishment	N.A.	Signaling prococols for LSPs set up LDP/TLDP [19], RSVP-TE [20]
Fault Detection	End-to-End detection	Local detection: BFD [50] End-to-End detection: LMP [53]
Path Recovery	Reactive discovery (Ponctual)	Local Protection FRR [67]

Table 3.1 LOCARN beside (G)MPLS mechanisms

that the transmission capacities provided by the current optical technologies (40Gbps and even 100Gbps) make this bandwidth consumption of non data packets (overhead) acceptable and even negligible in many cases – that's what we show in chapter 5.

LOCARN benefits as a Transport Technology

Immediate communication establishment The "on demand" establishment of a client communication between two edge nodes is almost instant because the Edge Nodes do not require any preliminary information about the network topology. Since two services identifiers are declared, the settling time simply depends on the Round Trip Time of the different collected paths (commonly around 20ms in national transport networks using FOs)
Since LOCARN discovers paths on demand rather than computing them, there is no convergence time, no complex calculation, no memory state to store and maintain like in proactive routing protocols (both LS or DV). Moreover, this swiftness allows LOCARN to perform purely reactive end-to-end path recoveries if failure are detected quickly.

Reactive path recovery requiring no configuration Traditionally, in transportation networks, the resiliency to server layers failures is obtained by the configuration of sub-paths protections – protection schemes 1 + 1 or 1:N. This approach, inherited from circuits networks, is applied in (G)MPLS architectures (FRR) because of the convergence time involved by IGPs (around tens of seconds). Clearly, this method is costly for the network operator. In terms of OPEX (configuration), in terms of CAPEX (when a line is specifically allocated), and in terms of management complexity. For example, the isolation of an equipment remains nowadays an heavy and risky operation.
On the contrary in LOCARN, the resiliency (and therefore the availability) of the network is considered through reactive mechanisms which by definition require no preliminary con-

figuration. Typically, LOCARN is able to recover a communication under 50ms if the path failure has been detected in about 30ms – 50ms is a commonly accepted threshold to define “Carrier Grade" networks.
Morevover, since the LOCARN recovery is very quick and since there is no intermediate forwarding state to maintain in nodes, the isolation of an intermediate node is quite simple: this latter juste have to be unpluged and all depending paths will be recovered instantly.

Adaptiveness to both client and server layers As mentioned before, each LOCARN service adapts its active path opportunistically according to the network recent state, including both the infrastructure topology (discovered paths) and the effective traffic loads.

For example, if the goal is to use at best the bandwidth provided by an infrastructure, then a service will choose the best path for being the “less used" one in terms of bandwidth occupancy rate. Concretely, a score must be associated to discovered paths on the base of information collected on paths interfaces’ statistics. Then, when a path is selected some client data traffic pass through it and consume bandwidth over time, making the path occupancy score less and less attractive for posterior services path selections. Thus, the succession of services path selections will tend to spread the client traffic within a domain in an opportunistic and best effort way.

In the perspective of an application to Transport Networks, a such self-adaptive behavior would allow network providers:

– A dramatic simplification of the network dimensioning process in comparision to approaches based on resources reservation where all must be planned and controled. LOCARN proposes a totally plug&play and on demand approach of the network resources management. The simple addition of « resources » in the underlying layer (node, link, bandwidth, queuing memory) among a network domain increases the *global network transmission capacity*. As well, without involving any configuration, the addition of nodes or links increases the potential paths for almost all services of a domain, and thus the *global network resiliency* and *availability*. Hence this plug-and-play oriented management of the network both suppresses the dimensioning process and allows to improve the Time To Market in case of new deployment or network expensions.
  Moreover, both resiliency and the adaptativeness are tunable in LOCARN. It can even be tuned specifically for each service for a fine configuration, in order to provide differentiated level of service for example.

- A typical transport network has 30% utilization; it must reserve bandwidth for “burst" times. The oppotunistic routing according to bandwidth should allow the network provider to minimize significantly the over-sizing of backbone networks by spreading loads during the pics of traffic.
- By including the available bandwidth in the routing policy criteria of path adaptation, LOCARN is able to *prevent congestion* (i.e. data transfers bottlenecks) by design, which both IP and IP/MPLS are unable to do.
- In an energy saving perspective, the LOCARN paths adaptiveness coupled with an adapted mechanism to turn off/turn on network devices would allow to adapt in real time the network electic consumption according to the effective needs.

LOCARN shortcomings as a Transport Technology

Overhead Costs In comparision to both IP and IP/MPLS intradomain protocols, the LOCARN overheads is clearly more important. In particular, the overhead become more and more important within a domain when the number of active services increase. Thus the scalability issue of the architecture due to its overhead is the primary challenge to be addressed.

Lost of control / design radicality LOCARN proposes a simple but radical way to operate transport networks that is in rupture with legacy approaches in those networks. Indeed, central idea of LOCARN is to let the control plane adapt/recover end-to-end paths, avoiding both heavy traffic engineering steps (computation / preliminary configuration) and admission control. A such approach can a priori fear a network operator because the traffics' end-to-end paths are not garanteed to be stable. In fact this is not a real problem but the necessary counterpart: if an operator wants to simply its network management, it necessarily goes with automation (self-* behaviors), it is to say with some loss of its own decision making power.

No VLANs In the design presented, LOCARN does not assume VLANs because we think it is not necessary and even not suitable. First, the use of VLANs involve some preliminary steps of configuration/planification which aim to be reduced the minimum in the LOCARN operational approach. Otherwise, the actual interests of VLANs in transport networks are the following:

- *switching nodes simplification and ability to scale (simple read of a VLAN label)*: in LOCARN, data relaying is achieved by *autoforwarding* that is already very simple

and scalable switching operations – next hop is just read from the data header, no forwarding state involved.

– *data traffics isolation*: in LOCARN, a service identifier field is present in data packets headers, which allows by design to garantee the data traffics isolation as well as a VLAN label

Nevertheless, if it becomes a requirement it is possible and even easy to assume VLANs in LOCARN in a similar way than the Ethernet “Max-in-Mac" specification (802.1ah, [44]) – it is only a datagram specification issue.

No Bandwidth Reservation In line with the “nethead" best effort paradigm, the LOCARN proposed design does not assume bandwidth reservation, it is to say that it does not assume some hard QoS garanteeds for its services. The reason of this choice is preserve a global consistency of the LOCARN proposal. Indeed, an architecture which continuously ajust its paths is much opportunistic if the not effectively used bandwidth of a domain is entirely available for potential routing changes. Yet, as well as for VLANs, the bandwidth reservation remains a possible and even a quite simple functionnality to introduce in LOCARN. An extended hybrid LOCARN design would allow the cohabitation of both “hard" and “soft services" in a same domain. De facto, the hard services would be less adaptive than soft ones and less elastic (shortest paths) whereas soft services would be still adjusted to the available bandwidth through the domain.

3.2.3 LOCARN as a Control & Management Proposal

Possible complementarity with SDN towards autonomic networking We consider that LOCARN’s good properties exposed previously make it a suitable base for the developement of a flexible, reactive, autonomic transport networks. Indeed, a transport network based on LOCARN would profitably include a centralized controler in charge of the global orchestration of the LOCARN layer by retroacting on its configuration parameters according to the observed network evolution. The paths discovery/selection can be adapted at the Edge Nodes (path selection rules, Optimization Interval (OI), path request TTL), as well as the *OAM settings* (Service Check Interval (SCI)). Hence, such a central controler would consider the end-to-end services as the managed resources by interacting with its Edge Nodes both to collect information and to modify on the LOCARN parameters according to a global planification following High Level Objectives. This way, it is altogether possible to build an autonomic network uppon a LOCARN layer. In particular, this centralized controler could

be a Software Defined Networking (SDN) controler, interacting with ENs with Openflow in a context of an unified integration of LOCARN to the SDN framework.

Beside the Autonomic Transport Networks paradigm According to the terminological distinction exposed in [60] (beginning of section II), we prefer the “automatic" term rather than the “autonomic" one to qualify LOCARN since in the architecture, no High Level Objectives (HLOs) are defined to pilot the path adaptations criteria. However, an Autonomic Network Management System (ANMS) could be introduced quite easily for the coordination or ENs decisions on the base of the information they have collected over time. This way, LOCARN can be the mean to introduce autonomic networking principles in an incremental way.

An opposite approach to PCE architectures Compared to intradomain PCE proposals, LOCARN involves a complete orthogonal approach of the network control.

The PCE proposals underlying paradigm is inherited from the traditional “bellheads" approach of operators. This latter assumes the centralization of the decision making through various kinds of NMSs (centers of automated control processes as well as entry point for the application of external Operations Support Systems (OSSs) instructions) aiming ultimatelly to determine and apply a globally optimal configuration.

On the contrary, LOCARN follows a best effort and distributed approach. In LOCARN the overall network configuration is adapting towards a suitable configuration, but since it does not aim to reach an optimal configuration, it does not have to be centralized or preliminary planed which systematically involves very complex constraint problems in large networks (commonly treated with methods and algorithms from the operational research field). Following a hollistic approach, the LOCARN architecture adapts the global network configuration by involving several, distributed and very simple control mechanisms.

3.3 LOCARN Modeling and Simulation

LOCARN Design and Simulation's Object Model The object model implemented in OMNeT++ followed the functional blocks exposed in the Fig. 3.4. Indeed, LOCARN can be decomposed into three kinds of functional blocks:

- the *Relay Unit (RU)* is in charge of LOCARN packets forwarding. Two types of forwarding processes: (i) point-to-point autoforwarding for relaying of data and OAM packets ; (ii) hop by hop duplications/forwarding for the diffusion of flooding packets. No state is involved in any of those processes.
- the *Control Unit (CU)* is in charge of services registration [11]. It is then responsible of the source routing associated to its services CORPs. Indeed, it triggers floodings, collects answers, selects the best path and (re)configure the concerned Edge Unit registers for incoming packets encapsulation. In charge of OAM exchanges, fault detection and path restoration process. It memorizes autoforwarding information and eventually some path meta-information.
- the *Edge Unit (EU)* is in charge of the DATA packets encapsulation / disencapsulation. It stores autoforwarding headers of the current path for each service – typically implemented with CAM registers.

Two kinds of nodes roles [12] :

- *Transit Nodes (TNs)*: Forward packets: process autoforwarding or flooding propagation (duplications). Solely contains a Relay Unit functional block.
- *Edges Nodes (ENs)*: ENs are ingress and/or egress nodes of a LOCARN domain. Services are defined by peers of reference points CORP/CDRP. Wether a service S, the S origin and destination ENs are respectively the nodes where the CORP and CDRP are declared. The S origin EN is in charge of the path establishment and maintenance to the destination. Edge Nodes contains both a Relay Unit, a Control Unit and an Edge Unit functional blocks.

Three kinds of transport entities, namely according to the G.805 denomination:

- *Link Connection (LC)*: a packet direct link between two Relay Unit upon which LOCARN frames are transmitted (i.e. into which those latter are encapsulated).

11. it may memorize services' specific SLS to set up some differentiated levels of services
12. In practice, a concrete LOCARN node can be at the same time an Edges Node (EN) for one service and a Transit Node (TN) for another one.

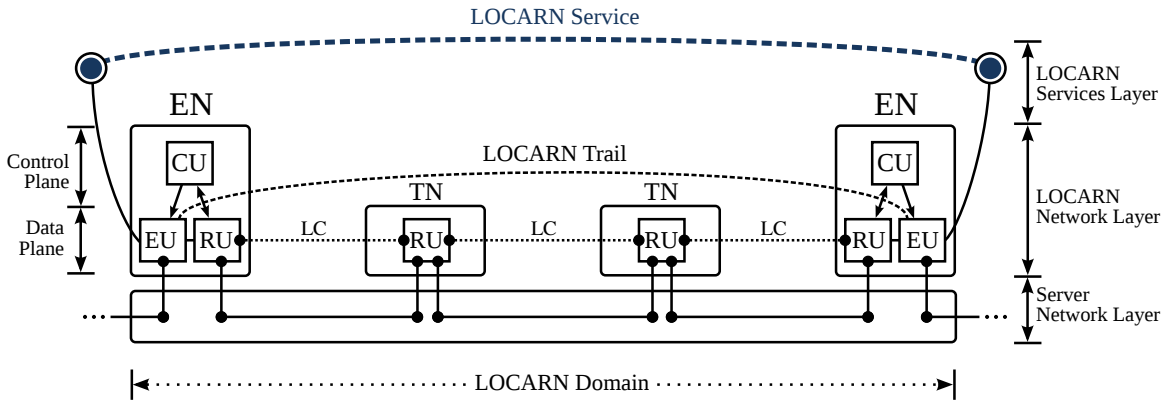


Figure 3.4 The LOCARN functional blocks

– *LOCARN Trail*: an end-to-end communication established from one point to another one through a LOCARN domain.
– *LOCARN Service*: a point-to-point transportation service provided to a client layer under the form of a couple of Access Ports (APs). A trail is established, maintained and adapted according to the service requirements and the evolution of the network state over time (see Fig.A.1).

About the Omnet++ Implementation The normal usage of the OMNeT++ framework is based on specifics `.ned` intantiation files. These latter describe the involved network components, their interconnections and settings. The framework also allow to configure some events to be triggered during the simulation processing. Thus, `.ned` file constitute the interface of interaction with the OMNeT++ framework and its discret event engine [13].

However the usage of this interface can be quite tedious since all interconnections between nodes must be decribed explicitely in the file. In our case we wanted to test LOCARN on numerous topologies by using specific input XML files containing both the network topology and services endpoints distribution. Thus, to increase the flexibility of the simulator and to accurately master the objects instantiation, we developped ou own routine (adapted to our XML files syntaxe) than use `.ned` files.

LOCARN behaviors The LOCARN main behaviors are depicted under sequence diagrams in annex, namely: (i) services establishment in Fig. A.2 ; (ii) service supervision A.3 ; (iii) service corouting synchronization in Fig. A.4 – assuming their bidirectional corouted

13. their are read at the beginning of the simulation for objects instantiation

services.

About Performances With a quad core CPU, the LOCARN simulations corresponding to the results presented in chapters 4 and 5 took less than one hour. The most impacting factors of the processing duration are the number of services and above all the TTL setting. Those two parameters are respectively responsible of the number and the magnitude of flooding processes, it is to say the amount of objects (packets) and associated events to process. The TTL setting is particularly decisive. Indeed, even for a single flooding propagation process, a TTL set upper to 11 hops makes the simulation duration exceeding one hour with a common topology density.

Chapter 4

Scalability and Performances Evaluation of LOCARN

As stated in the previous chapter, the critical issues of LOCARN are related to its ability to scale because the overheads produced. In this chapter, we study and quantify the LOCARN overhead production in order to prove the viability of the architecture. In section 4.1, we study its general performances and scalability in normal conditions whereas in section 4.2 we observe performances in cases of failures. The results of this section have been published in [56] and [57].

4.1 Overhead and Performances in Normal Conditions

4.1.1 Analytic Formula for the Floodings' Production of Messages

The two parameters determining the message generation of a flooding process bounded by a Time To Live (TTL) are the network density (δ_{net}) and the TTL value. Whether a network having N nodes and L links, the mean density corresponds to $\delta_{net} = L \times 2/N$ (the mean number of interfaces per node). The average message generation can be estimated quite easily: the origin node sends on average δ_{net} messages (across all interfaces), whereas successive nodes retransmit on average $\delta_{net} - 1$ until the TTL is reached (across all interfaces except the incoming one). Thereby, whether M_N representing the number of messages generated by duplications at the propagation step N, we have:

$$M_N = \delta_{net} \times (\delta_{net} - 1)^{N-1}$$

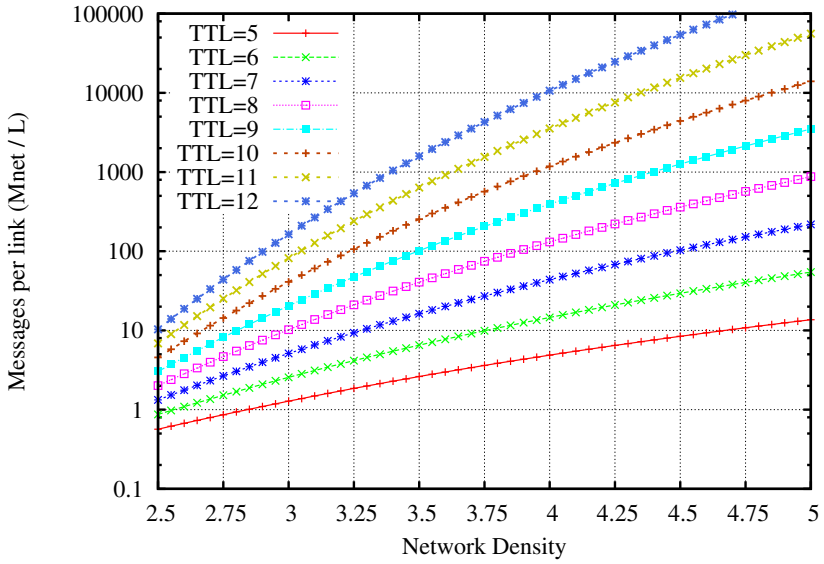


Figure 4.1 Evaluation of the message generation for a single flood according to the network density and TTL bound (reduced in messages per link)

Then we get an estimation of the global amount of messages generated over the network (M_{flood}) by summing terms from M_1 to M_{TTL}:

$$\begin{aligned} M_{flood}(\delta_{net}, TTL) &\approx \delta_{net} \times \sum_{i=1}^{TTL} (\delta_{net}-1)^{i-1} \\ &\approx \delta_{net} \times \frac{(\delta_{net}-1)^{TTL}-1}{\delta_{net}-2} \end{aligned} \tag{4.1}$$

This expression provides an *upper bound* estimation in so far it does not include that, along the flooding propagation, messages which achieved a cycle are discarded. Hence all messages estimated subsequently to loop detections are counted in excess, which yet allows to express the message generation *magnitude* through a simple formula. In Fig.4.1 are gathered numerical applications giving an overview of this magnitude reduced per link, by varying TTL (ranging from 5 to 12) and δ_{net} (ranging from 2.5 to 5). These results correspond to the numerical application of the formula 4.1 by considering a network with $N = 50$ nodes [1], dividing the global amount of messages generated over the network by the number of links ($L = (\delta_{net} \times N)/2$). Unsurprisingly the amount of messages quickly explodes with the network density, making the overhead cost unacceptable for “dense networks" and “excessive" TTL range.

1. application for a 100 nodes with the same density gives similar magnitudes, globally low

	Topological Description	Density (δ_{net})	Diameter (D)
Profile 1	“Scarce Extended"	2.8	10
Profile 2	“Medium"	3.5	7
Profile 3	“Dense Little"	5	5
Profile 4	“Medium Extended"	3.5	10
Profile 5	“Dense Medium"	5	7
Profile 6	“Dense Extended"	5	10

Table 4.1 Topological profiles parameters for analytic estimations

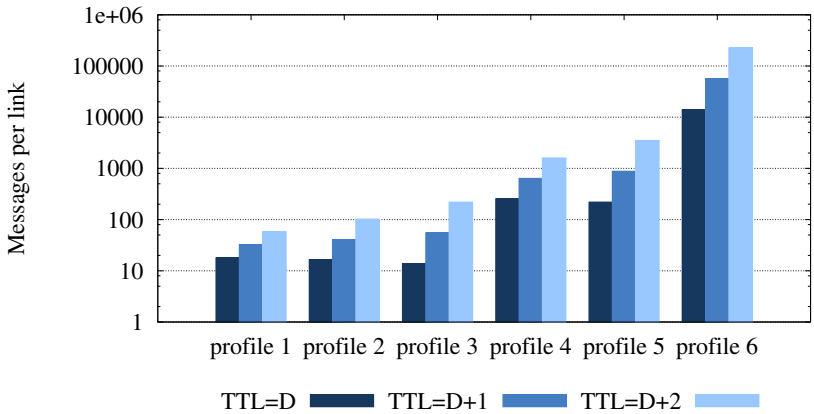


Figure 4.2 Message generation for a single flood within several network profiles and TTL arrangements

In Fig.4.2 are reported the magnitudes of the messages generation for one flood that can be expected per link upon several networks (profiles are exposed in Table 4.1). The TTL of flooding is chosen to exceed the diameter in order to estimate the most expensive service path discovery. The obtained results bring out that upon “reasonable" network dimensions (profiles 1, 2 and 3) a link transmits on average between ten and hundred of path request messages within a flooding process.

4.1.2 Path Discovery Results: Performances beside Overheads

In this section, we expose some LOCARN’s simulation results. The architecture mechanisms (data plane, control plane and OAM) have been implemented in the Omnet++ discrete event simulator.

A network infrastructure is simulated by taking into account: the network topology, the links datarates and the propagation delays (packet-processing delays are not implemented here as they are negligible by report to the propagation ones in the example taken, see [73]).

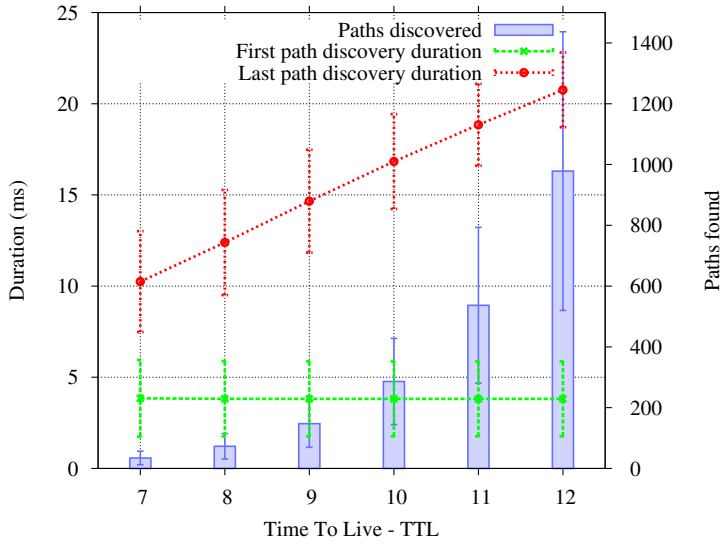


Figure 4.3 Path discovery performances (average and standard deviation for 1000 randomly picked services)

To study performances and costs over a concrete and realistic use case as those encountered in Orange Group, the results of this section correspond to a LOCARN application to the national (France) core network topology. The latter includes N=42 nodes, L=68 links, has a diameter D=7 hops and a mean density $\delta_{net} = 3.23$ whereas the nodes degree standard deviation is $\sigma_\delta = 1.12$. Propagation delays are implemented in accordance with the optical propagation across the real inter-nodes distances (the mean propagation delay is $445.9\mu s$). Nodes interfaces are associated with unlimited FIFO queues without priorization between data, control or OAM packets.

Path Discovery Performances and Overhead Cost

Hereafter we bring out the performances and overhead cost of the path discovery process, namely the number of paths discovered and the discovery duration beside the amount of overhead generated by the flooding discovery.

To get a statistical overview of the path discovery performances (Fig.4.3), we pick up 1000 services randomly among the topology: one random node as origin and a distinct random node as destination. Then we launch discoveries from each origin, varying TTL from 7 to 12 (which corresponds here to TTL=D and TTL=D+5). The answer packets arrival to the origin are counted and their arrival time is noticed to respectivelly get the number of paths discovered and the discoveries durations. Finally, the figure provides the first and last

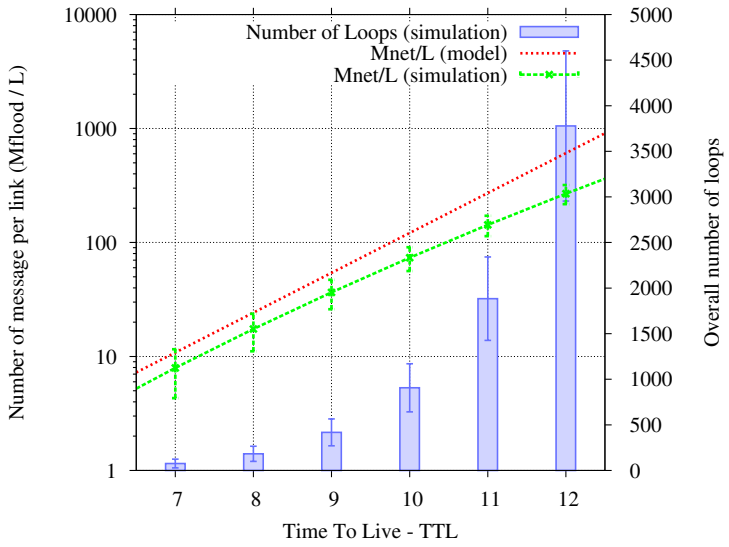


Figure 4.4 Messages generated per link for a single flood: model and simulation

path discovery duration through the average and standard deviation (respectively through green and red lines), refer to the left axis for durations. The discovered paths are expressed by histograms (read on the right axis). We must also note that the significant standard deviations (both for durations and number of paths) is due to the fact that the random picking of services give a significant variance of enpoints “remotess" which is itself determining on the path discoveries.

To get a statistical overview of the path discovery overhead cost (Fig.4.4), we launch this time one path discovery process for each of the 42 nodes and observe each time the message generation, varying TTL from 7 to 12. The overall generated messages during each flood (i.e. the request messages) are counted. Finally the figure reports the average number of flood messages reduced per link (green line); the previous analytic expectation (see section III.A) is reported for comparison (red line). The simulation confirms that the analytic formula provides an upper bound estimation which is relatively accurate for this network (due to the low standard deviation of nodes degrees, $\sigma_\delta = 1.12$). Yet, it is less accurate when the TTL deviates too much from the network diameter because the amount and size of loops become too significant to be ignored in the analysis. The number of detected loops is reported with the histograms (read on the right axis).

By crossing the Fig.4.3 and Fig.4.4 results, it is possible to bring out the relevance of TTL settings on the simulated network. For example with TTL=10, path discoveries provide on average hundreds of paths. If we assume for example that a such TTL setting gives sufficient

Network	TTL	S	OI	SCI	Floods overheads	OAM overheads	Cumulated overheads	Ratio 1Gb/s
Profile 1	12	100	10s	10ms	537Kbps	310Kbps	847Kbps	**0.08%**
				1s		3Kbps	540Kbps	**0.05%**
			30s	10ms	179Kbps	310Kbps	489Kbps	**0.05%**
				1s		3Kbps	182Kbps	**0.02%**
		1000	10s	10ms	5367Kbps	3108Kbps	8475Kbps	**0.85%**
				1s		31Kbps	5398Kbps	**0.54%**
			30s	10ms	1789Kbps	3108Kbps	4897Kbps	**0.49%**
				1s		31Kbps	1820Kbps	**0.18%**
Profile 2	9	100	10s	10ms	212Kbps	1764Kbps	1976Kbps	**0.20%**
				1s		18Kbps	230Kbps	**0.02%**
			30s	10ms	71Kbps	1764Kbps	1835Kbps	**0.18%**
				1s		18Kbps	89Kbps	**0.00%**
		1000	10s	10ms	2116Kbps	17644Kbps	19760Kbps	**1.98%**
				1s		176Kbps	2292Kbps	**0.23%**
			30s	10ms	705Kbps	17644Kbps	18349Kbps	**1.83%**
				1s		176Kbps	881Kbps	**0.09%**
Profile 3	7	100	10s	10ms	1328Kbps	169Kbps	1497Kbps	**0.15%**
				1s		2Kbps	1330Kbps	**0.13%**
			30s	10ms	443Kbps	169Kbps	612Kbps	**0.06%**
				1s		2Kbps	445Kbps	**0.04%**
		1000	10s	10ms	13284Kbps	1688Kbps	14972Kbps	**1.50%**
				1s		3Kbps	540Kbps	**0.05%**
			30s	10ms	4428Kbps	1688Kbps	5916Kbps	**0.59%**
				1s		3Kbps	4431Kbps	**0.44%**

Table 4.2 Analytic estimations of overheads due to non-data packets (per link among three network profiles)

amount of paths for the expected potential of services adaptation and recovery under failure. By looking at the discoveries durations, we see that the last path is obtained around 17ms – whereas the first path is still obtained around 5ms independently from the TTL. Under such discovery durations, a sub 50ms recovery of a broken path can be achieved if faults can be detected between 30ms and 45ms (typically, OAM should be sent every 10ms and a fault considered upon the lake of three packets). On the other hand, in terms of overhead cost, with TTL=10 each flood generates on average almost 70 packets per link.

Global Overheads Estimation Over Three Network Profiles

As exposed before, services involve floods for their establishment and periodically for their possible adaptations, while active paths are continuously checked by OAM exchanges. We call Optimization Interval (OI) the interval between path optimizations, and Service Check Interval (SCI) the interval between OAM packets exchanges. To estimate the OAM overhead, we have to consider the amount of active services (S), and the SCI interval. Yet the OAM message generation also depends on the active paths lengths: for example if all services use ten hops paths, the OAM exhanges from end-to-end will consume twice of the network resources as if they would use five hops paths. Hence, the exact overhead generated by OAM exchanges among a period of time actually depends on the active paths

lengths during this period. Hence, to get an estimation of the OAM overhead for S services, we have to make significant assumptions about the active path lengths. For a meaningful estimation, we assumed that the services' endpoints "remoteness" (i.e. their shortest paths in number of hops) distribution follows a gaussian distribution probability from 1 to D hops. Then, the effective selected paths can vary over time among the shortest path and the TTL value according to the successive optimizations in the paths selections. By observing that LOCARN active paths lengths rarely exceed the shortest path over two hops through simulations of a realistic transport network topology and traffic matrix, we are able to give an upper bound estimation of OAM messages per second and per link according to S and SCI.

We summarize in Table 4.2 the estimations of LOCARN overheads due to non-data packets, gathering estimations of floods overhead based on the III.A formula and estimations of OAM overhead. Results are reported for three network topological profiles. For each profile, the LOCARN settings (TTL, OI, SCI) are declined together with the amount of active services across the network. The TTL values are arranged to exceed the network diameter ($TTL = D+1$, $TTL = D+2$) providing an important potential for paths optimization. S, OI and SCI are chosen to expose orders of magnitudes ($S = 100/1000$; $OI = 10/30s$; $SCI = 10/100ms$). The last column reports the sum of overheads estimations per link in ratio beside 1Gb/s that is used here as an indicative basis datarate which allow transpose estimations for higher granularities. The numerical applications confirm that in absolute, the LOCARN overhead due to non-data packets is substancial: on average several mega bit per second and per link for a thousand services. However, in a Transport Network context, such overheads represent relatively few bandwidth consumption. At most, for a thousand of highly resilient ($SCI = 10ms$) and a highly adaptive ($OI = 10s$), the cumulated overheads reach on average 0.85%, 1.98% and 1.50% of 1Gb/s links respectively for profile 1, 2 and 3. Given that nowadays the Transport Networks transmission capacities are commonly around 10, 40 or 100Gb/s in core segments, a such overhead magnitude remains quite acceptable (notwithstanding, the bandwidth consumption estimated assume the packets transfer in the two links directions, hence the effective consumption on average corresponds actually to half of the last column).

Finally, these analytic results permit to point out that the real scalability issue of LOCARN in terms of overheads' mean bandwidth consumption in a transport network context relies on the fact that both for control and OAM, the non-data packets generation linearly depends

on *the amount of services active in the network (S)*. This is clearly the consequence of the “service oriented" LOCARN design: each service involves its own generation of packets both for routing and OAM.

4.2 Overhead and Performances Over Failures

As observed in the previous section, floodings due to path discoveries produce numerous packets across the network consuming the network bandwidth accordingly. In section 3 we have thus studied the amount of packets generated, and finally express the overhead cost in terms of mean bandwidth consumption over time. In subsection 3.3, overhead estimations assumed “normal conditions", it is to say that only path optimizations’ floodings and OAM exchanges were considered. However, the bandwidth consumption’s means values over time (like in Table 4.2) do not account that floodings occur along very short durations actually depending on the infrastructure propagation delays. Indeed, the floodings do not consume the bandwidth constantly over time, but produce consumption’s peaks that must be monitored. Considering the previous section example, if a flooding involves on average 70 packets per link in both two directions (see Fig 4.4, $TTL = 10$), thus interfaces belonging to the flooding domain will receive around 35 packets during 17ms, which is equivalent to 2060 packets/s.

In addition, the potential problem in LOCARN is that several floodings processes can overlap (i.e. serval flooding processes can be in progress at the same time) which leads to cumulate the peaks of overheads. In normal conditions, given that floodings are launched periodically for each service (“path optimizations"). Given the order of magnitude forseen for optimization intervals in comparison to the floodings duration, the amount of overlapping discovery processes is statistically very low. However, when an infrastructure failure occurs (node or link), if S services are disrupted, S discovery processes are triggered from all the S services source nodes. If the detection time is tuned to very short times (few milliseconds), then the probability of overlapping becomes very high. The overlapping of the S floodings messages generation may cause the filling of packets queues, and even impact the Quality of Service (QoS) of data transmission, in particular jitter[2]. Yet the impact of floods overlapping is difficult to estimate in a comprehensive way because it may vary a lot according to: (i) the spreading of floods starting over time; (ii) links’ propagation delays; (iii) links’ available bandwidths during floods propagation.

2. as we consider a same priority level both for data and control packets relaying for the sake of simplicty

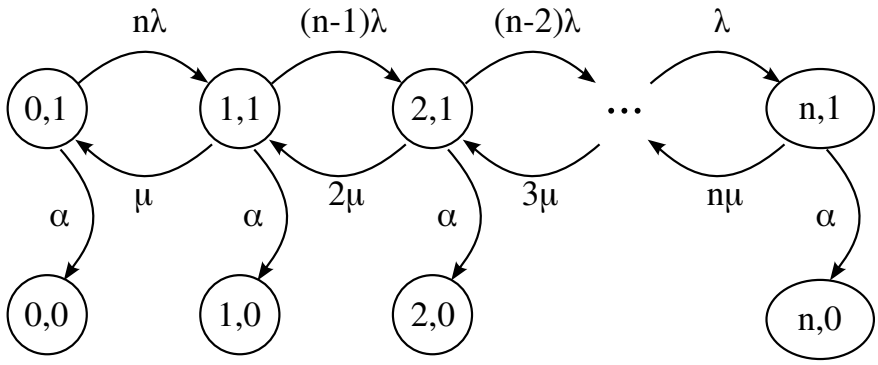


Figure 4.5 The markov transition diagram – Evolution of services depending on one trail and the risk of its failure

In this section, we study at first the path recovery that can be expected statistically according to the infrastructure reliability to get a general overview of overlapping floodings according to failures probability. Secondly, we observe by simulations the maximum impact that overlapping recovery processes may involve on the data plane transmission to get an overview of the worst effects of failures.

4.2.1 Path Recoveries Statistical Expectation

To apprehend in a general way the services path recoveries statistical impact, we propose an analytical model to derive the number of services affected by a path failure in LOCARN. Focusing on the path level permits to simplify the modeling process without losing generalities, and hence obtain a trend of the overhead's impacts.

In the envisioned model, we assume that: (i) the inter-arrival time of each service follows an exponential distribution with rate λ; (ii) the service duration follows an exponential distribution with rate μ. In the same way, the time duration before a path failure occurs is exponentially distributed with rate α. We consider that the maximum number of services supported by a LOCARN path is n. The above assumptions conduct us to model the system using a Markov Chain $X = \{X_t, t \geq 0\}$ on the state space S defined by $S = \{(i,k) | i = 0,\ldots,n, \text{ and } k = 0,1\}$, for every $n \geq 1$. $X = (i,j)$ means that, at time t, there are i active services in the path and the latter is in state k. $k = 1$ indicates that the path is ok, while $k = 0$ means that the path is in failure state. Fig. 4.5 shows the transitions graph of the system. We remark that all states where $k = 0$ (failure) are absorbing states. Such design permits to know the number of active services when the system enters an absorbing state. Here, we focus only on capturing the system state when it fails. The case of reestablishing the path connection is out of scope of this work.

The different transitions of this chain are as follows:

- If a service arrives while already i ($0 \leq i \leq n-1$) services are active and the path is ok, then there is a transition from state $(i,1)$ to $(i+1,1)$ with rate $(n-i)\lambda$.
- If a service leaves while already i ($0 \leq i \leq n-1$) services are active and the path is ok, then there is a transition from state $(i,1)$ to $(i-1,1)$ with rate $(i)\mu$.
- If a path failure occurs while already i ($0 \leq i \leq n-1$) services are active then there is a transition from state $(i,1)$ to $(i,0)$.

We denote by Q_B the transition matrix between non-absorbing states. It is worth noting that this matrix does not represent the infinitesimal generator of this chain. Let σ_B be the initial probability distribution vector of the chain states, and $Q_{B,k}$ the vector containing the transitions rate from the non-absorbing states to the absorbing state i. Q and $Q_{B,1}$ could be obtained as follows:

$$Q_T = \begin{bmatrix} -(n\lambda+\alpha) & n\lambda & 0 & 0 & \dots & 0 \\ \mu & -((n-1)\lambda+\alpha+\mu) & (n-1)\lambda & 0 & \dots & 0 \\ 0 & 2\mu & -((n-2)\lambda+\alpha+2\mu) & (n-2)\lambda & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \dots \\ \vdots & \vdots & \vdots & \vdots & n\mu & -(\alpha+n\mu) \end{bmatrix}$$

$$Q_{B,1} = \begin{bmatrix} \alpha \\ 0 \\ 0 \\ \vdots \\ \vdots \end{bmatrix}$$

The probability π_k to be in the absorbing state k (i.e. the state (k-1,0) of our notation)) is obtained as follows:

$$\pi_{k-1,0} = -\sigma_B (Q_B)^{-1} Q_{B,k}$$

We note the vector v as $v = -\sigma_B (Q_B)^{-1}$. Knowing that the initial state from which the system begins is state (0,1) and by deduction we can obtain the linear system $vQ = -\sigma_B$ that can be written as follows:

$$\begin{cases} -(n\lambda+\alpha)v_0+\mu v_1=-1 \\ (n-i+2)\lambda v_{i-2}-((n-i+1)\lambda+(i-1)\mu+\alpha)v_{i-1}+i\mu v_i=0 \quad for\ i=2\ to\ n \\ n\mu v_{n-1}-(n\mu+\alpha)v_n=0 \end{cases}$$

Knowing that $\pi_k=\frac{v_k}{\alpha}$ (since $\pi_k=vQ_{B,k}$) and replacing its value in the precedent linear system we obtain:

$$\begin{cases} -(n\lambda+\alpha)\pi_0+\mu\pi_1=-\alpha \\ (n-i+2)\lambda\pi_{i-2}-((n-i+1)\lambda+(i-1)\mu+\alpha)\pi_{i-1}+i\mu\pi_i=0 \quad for\ i=2\ to\ n \\ n\mu\pi_{n-1}-(n\mu+\alpha)\pi_n=0 \end{cases}$$

We replace the last equation by the normalizing condition $\pi\mathbb{1}=1$ we obtain:

$$\begin{cases} -(n\lambda+\alpha)\pi_0+\mu\pi_1=-\alpha \\ (n-i+2)\lambda\pi_{i-2}-((n-i+1)\lambda+(i-1)\mu+\alpha)\pi_{i-1}+i\mu\pi_i=0 \quad for\ i=2\ to\ n \\ \pi_1+\pi_2+\pi_3+\cdots+\pi_n=0 \end{cases}$$

Accordingly we have the following recurrence relation:

$$\begin{cases} \pi_1=\frac{(n\lambda+\alpha)}{\mu}\pi_0-\frac{\alpha}{\mu} \\ \pi_i=\frac{((n-i+1)\lambda+(i-1)\mu+\alpha)}{i\mu}\pi_{i-1}+\frac{(n-i+2)\lambda}{i\mu}\pi_{i-2} \quad for\ i=2\ to\ n \end{cases}$$

To solve this recurrence, we start with any positive value of π_0, and then we calculate all values of π_i (for $i=1,\ldots,n$). Then, we obtain the real values of π_i by dividing each value by the sum of π_i. Having calculated the different probabilities (π_i), we are able to compute the expected number of active services ($E[S]$) on the path when a failure occurs:

$$E[S]=\sum_{i=0}^{n+1}(i-1)\pi_{(i-1,0)} \tag{4.2}$$

In fact, $E[S]$ gives the expected number of active services in the path when a failure occurs. Knowing that each service generates M messages in order to find another path, the expected number of messages generated in this case $E[M]$ is:

$$E[M]=ME[S] \tag{4.3}$$

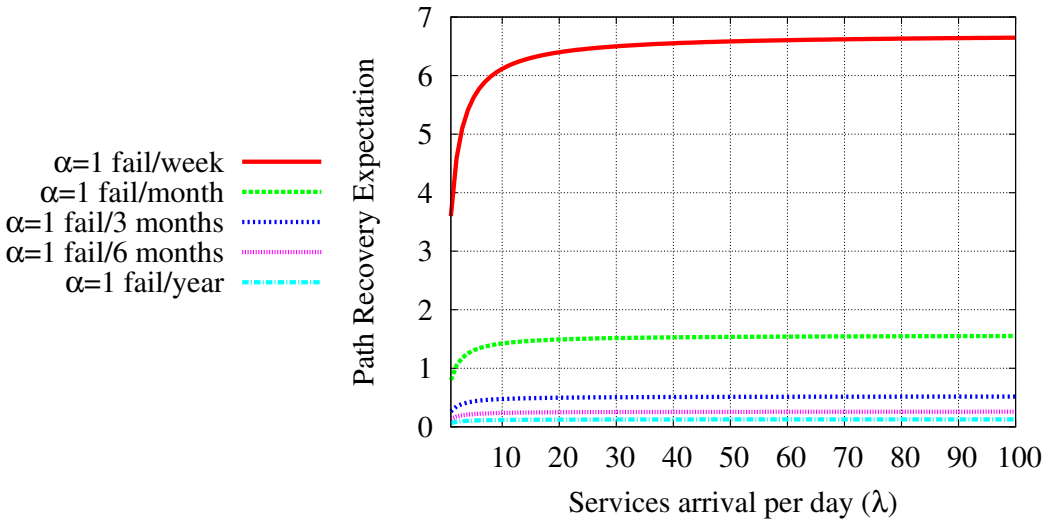


Figure 4.6 Expectation of services disconnections per day due to the path failure ($n = 50$)

Usually, we are interested on the number of messages generated during a period of time. We know that the mean time before absorption (i.e. failure) of the modeled system is $\frac{1}{\alpha}$. Therefore, the expected number of messages generated during a period (say day, if α represents a failure rate by day) is equal to:

$$E[MPeriod] = \alpha E[M] \tag{4.4}$$

Results and Observations

At this point, we are able to get the number of recovery processes that can be expected when λ, μ and α are known with a maximum of n fixed services by using (4.2). All the results thereafter assume that services mean duration before leaving is on average $\frac{1}{\mu} = one\ day$, knowing that what is finally impacting is the ratio $\frac{\lambda}{\mu}$. In Fig. 4.6, by using realistic transition rates magnitudes, we provide the expectation of services recovery processes launched per day ($E[S/day] = \alpha E[S]$) for $n = 50$ in Fig. 4.6. Once the maximum amount of maximum services n is reached by the markov process, the services disconnections expected almost only depends on n and α. It can be observed that for the choosen α and n values, the ratio $\frac{\lambda}{\mu}$ has a noticeable impact on expectation until 10. Those permit to see that, upon realistic settings, the significant impacting factors are α and n. To get a larger overview, some result are gathered in Table 4.3 where the same observations can be made.

To exhibit more convenient values, it is possible to apply the analytic formula (4.1) from section 3.1 to determine M, which allows us to find finally the messages expected

	$\lambda =$ \ $\alpha =$	$\frac{1}{year}$	$\frac{1}{6months}$	$\frac{1}{3months}$	$\frac{1}{month}$	$\frac{1}{week}$
$n = 20$	$\frac{10}{day}$	0.042	0.086	0.172	0.516	2.215
	$\frac{30}{day}$	0.045	0.091	0.183	0.548	2.351
	$\frac{100}{day}$	0.046	0.093	0.187	0.561	2.395
$n = 50$	$\frac{10}{day}$	0.117	0.237	0.475	1.425	6.116
	$\frac{30}{day}$	0.125	0.253	0.505	1.516	6.500
	$\frac{100}{day}$	0.127	0.259	0.517	1.551	6.647
$n = 100$	$\frac{10}{day}$	0.242	0.490	0.980	2.941	12.618
	$\frac{30}{day}$	0.257	0.522	1.043	3.129	13.414
	$\frac{100}{day}$	0.263	0.534	1.067	3.201	13.721

Table 4.3 Amount of paths recovery launches expected (per day)

	$\lambda =$ \ $\alpha =$	$\frac{1}{year}$	$\frac{1}{6months}$	$\frac{1}{3months}$	$\frac{1}{month}$	$\frac{1}{week}$
$n = 20$	$\frac{10}{day}$	1.722	3.492	6.984	20.964	90.075
	$\frac{30}{day}$	1.833	3.717	7.433	22.300	95.580
	$\frac{100}{day}$	1.875	3.802	7.602	22.794	97.390
$n = 50$	$\frac{10}{day}$	4.760	9.653	19.308	57.941	248.694
	$\frac{30}{day}$	5.067	10.275	20.551	61.656	264.322
	$\frac{100}{day}$	5.184	10.512	21.025	63.074	270.304
$n = 100$	$\frac{10}{day}$	9.824	19.922	39.847	119.570	513.065
	$\frac{30}{day}$	10.458	21.206	42.413	127.244	545.445
	$\frac{100}{day}$	10.699	21.696	43.392	130.176	557.917

Table 4.4 Amount of packets expected (per link and per day)

per link and per day by considering realistic network dimensions and LOCARN settings (i.e. δ_{net}, D and floodings' TTL). Doing so, we provide in Table 4.4 some estimations of path recoveries overhead that can be expected, based on both realistic transition rates and network dimensions. Network dimensions assumed are the same than in the section 3.2 simulations, we assume the France optical national backbone (i.e. N=42 nodes, L=68 links, diameter D=7 hops, the mean density $\delta_{net} = 3.23$ and the nodes degree standard deviation is $\sigma_\delta = 1.12$). Hence, to get a consistant view for transport networks estimation, we can specifically look at results with $\alpha = \frac{1}{year}$ or $\alpha = \frac{1}{6months}$. Here we see that in terms of packet expectation per day, the path recovery overheads are unsignificant since the maximum of services n is acceptable.

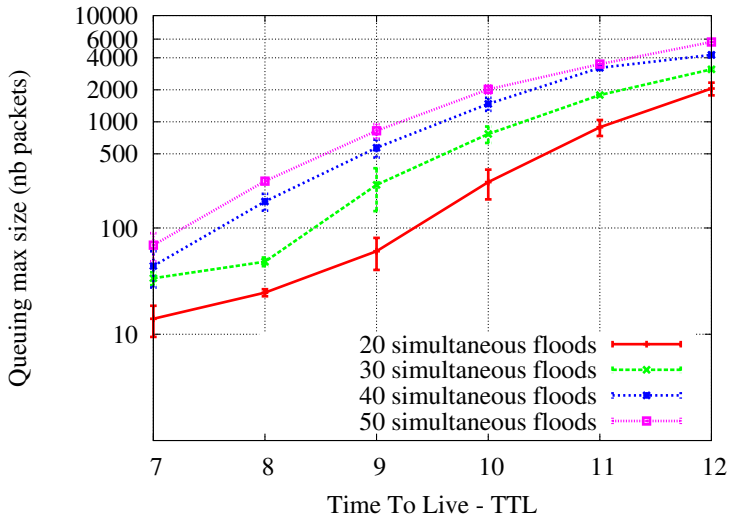


Figure 4.7 Queue maximum length along the studied path for N simultaneous floods over the network (1Gb/s links)

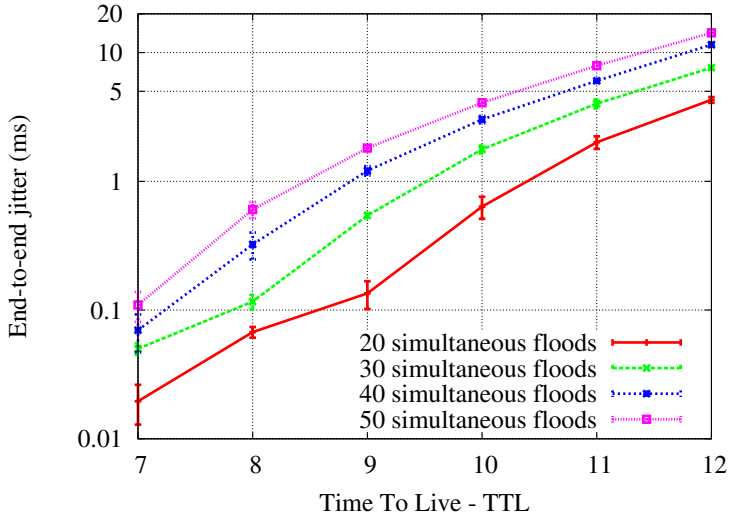


Figure 4.8 Worst data packet jitter due to additionnal queuing loads along the studied path for N simultaneous floods over the network (1Gb/s links)

4.2.2 Path Recovery Maximum Impact

As we have seen, floods due to path discoveries generate numerous packets across the network, consuming bandwidth. Until now we have studied the number of packets generated, and the involved overhead cost in terms of its per link and per flood bandwidth global consumption over time (Table 4.2). Yet, beyond the amount of packet generated, the critical aspect of floods' message generation is to occur along a very short period of time (which is

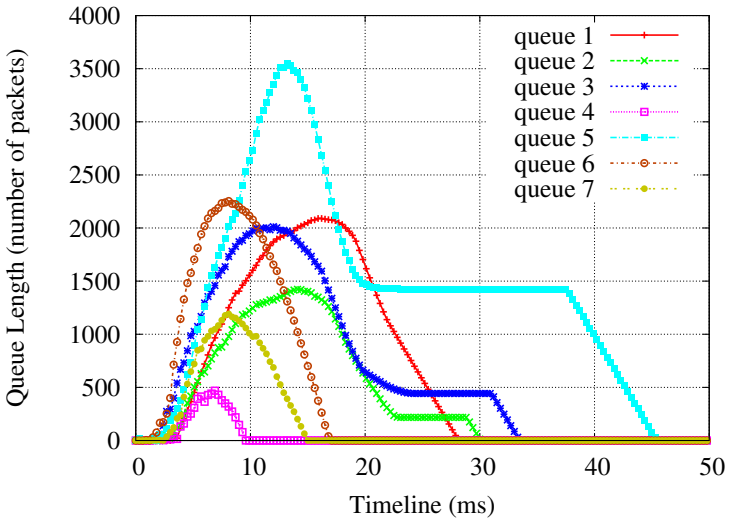


Figure 4.9 Evolution of queing load along the studied path for 1Gb/s links, N=42 simultaneous path discoveries, TTL=12

actually related to the infrastructure's propagation delays). Consequently, flooding does not constitute regular overheads but produce peaks of bandwidth consumption. Following the previous subsection example, if a flooding with $TTL = 10$ involves on average 70 packets to be transfered per link (i.e. in the two direction); thus an interface belonging to the flooding domain will have to cope on average with the reception of 35 packets during 17ms, i.e. almost 2060packets/s during this period. Moreover, overhead peaks may cumulate if several path discoveries are overlapped (i.e. occur in the same time).

In normal conditions, a flooding process is launched periodically for path optimization and statistically the amount of overlapping discoveries is verly low by considering the ratio of Optimization Interval (OI) to the floodings duration. Yet, in case of failure (node or link), when N services are disrupted, N discovery processes are triggered across the network from the disrupted services' origin nodes. If the detection time is tuned very quick, the probability of overlapping becomes very high. The overlapping of the N floodings messages generation may cause the filling of packets queues, and even impact the Quality Of Service (QoS) of data transmission, in particular jitter[3]. Yet the impact of floods overlapping is difficult to estimate in a comprehensive way because it may vary a lot according to: (i) the spreading of floods starting over time; (ii) links' propagation delays; (iii) links' available bandwidths during floods propagation.

3. as we consider a same priority level both for data and control packets relaying for the sake of simplicty

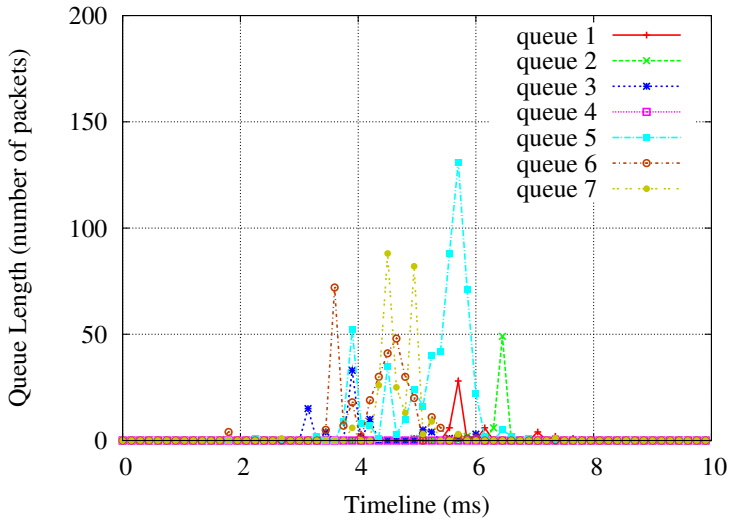


Figure 4.10 Same scenario than Fig.4.9 with 10Gb/s links

With Fig.4.7 and Fig.4.8, we exhibit the impact of N overlapping floods in terms of queues load and jitter. To bring out some meaningful estimations, we consider a very worst case scenario, providing the very maximum impact that can be expected on the network involving N overlapped discovery proccesses with different TTL. We proceed as follows: all links capacities are set to 1Gb/s (that we use as an indicator like in Table II). A constant data traffic is sent over a path “especially at risk" [4] in the network up to two-third of capacities. Then we launch *at the same time* [5] N discovery processes (N=20,30,40,50) by varying TTL from 7 to 12. We report in Fig.4.7 the maximum queue length that have been reached all along the studied path, until the end of flooding sessions and in Fig.4.8 the worst packet end-to-end jitter [6] along the data path. To bring out impacts independently from the starting points, we launch 20 simulations involving randomly picked origins samples and report the averages and standard deviations.

Finally in Fig.4.9 and Fig.4.10 we observe the evolution of queuing length over time along the previous sensible path in the case all nodes (42) would launch a flood at the same time, in case of recovery for example. Figures depict the evolution of queues lengths over time along the studied path. Respectively Fig.4.9 and Fig.4.10 permit to show how much

4. such a path have been identified as exposed to flooding perturbations over several simulations: it is a long path passing through the dense network zone

5. the fact than N floods are launched strictly at the same time actually corresponds to a very worst case that statistically never happens. This allow us to bring out an upper bound of impacts on the network.

6. by substracting arrival from sending times we get all packets delays. The worst packet jitter is simply obtain in the simulator by substracting the longest delay reached over the simulation by the shortest one.

the impact is reduced by passing from 1Gb/s to 10Gb/s links.

Chapter 5

Towards a Large Scale LOCARN Design

5.1 Motivation and Principles of the Two Proposals

5.1.1 Motivations

In the previous chapter, we made several observations related to LOCARN's performances and scalability. Concerning the routing plane, we explained that the counterparty of the hollistic path distributions/adaptations was the amount of overhead due to the periodical floodings. Globally, the overhead depends on first, the magnitude of one flood (which is a function of the network topology and the flooding Time To Live) and secondly the interval between floods. The impact of those three factors has been extensively studied in [56]. Concerning the OAM, we explained that the ability to recover upon a path disruption in few milliseconds and in a purely reactive approach is also possible to the price of a non negligible overhead cost because of the very frequent end-to-end exchanges.

The initial LOCARN design, presented in chapter 4 and evaluated in chapter 5 is "*service oriented*", it is to say that the three basic mechanisms (autoforwarding, enhanced flooding, end-to-end fault detection) are built around the service functional entity. In particular, source routing and OAM fault detection are operated for each service. This is a big difference with usual packet-based network standards that consider at least a node level (for routing or path recovery). Based on this observation, we describe in the next section two proposals that aim to de-correlate the control plane and OAM overheads from the number of declared services while maintaing the per service path determination. To do so, we have to introduce a quite more complex mechanisms in both Edge and Transit Nodes.

5.1.2 Principles of the two proposals

First Proposal: Multi-Services Path Discovery Our first proposal is simple but efficient. In the “service oriented" routing approach of the initial LOCARN design, if S services origin (CORPs) are declared on the same node, S path requests are sent for the connectivity establishment, but – above all – S path requests are periodically sent for path optimizations. To reduce the overhead, we observe that a single flooding could be launched for the discovery of all paths toward the desired destinations. Instead of launching S flooding requests that looks like “i’m looking for the destination of service A", an Edge Nodes can flood a message looking like “i’am looking for destinations of service A, B, C, ..." and thus collects all answers at the same time.

This Multi-services request approach has two consequences: (i) first, it drastically decorrelates the amount of generated request packets from the number of active services on the domain – what we were looking for; (ii) secondly, it increases the size of request packets due to the concatenation of multiple service identifiers. Globally, the correlation between the flooding overhead and the service remains but the number of request packetdrastically reduces when the amount of services per Edge Node increases. Finally, let’s notice that concerning the routing behavior, this solution should give at first sight less fineness than the initial “service oriented routing". A trade off must be found between the level of fineness and the global overhead gain expected.

In terms of implementation, the request packets (that are flooded across the domain until the destinations) now carry a list of service identifiers. When a destination node (EN) receives such a packet, it returns as many response packets as number of service identifiers from the list matching with service identifiers registered on this destination node.

Second Proposal: Point-to-multipoint Autoforwarding Our second proposal is quite more complex. As stated before, the initial LOCARN architecture uses “autoforwarding". It means that a packet includes in its header a sequence of ports, allowing each transit node to directly read and switch the packet along its path – a cursor is memorized and incremented at each hop to pointer directly to the overhead part.

In practice, autoforwarding is used in LOCARN for the end-to-end transmission of Data and OAM packets; it constitutes a strong point of the design because it is at the same time

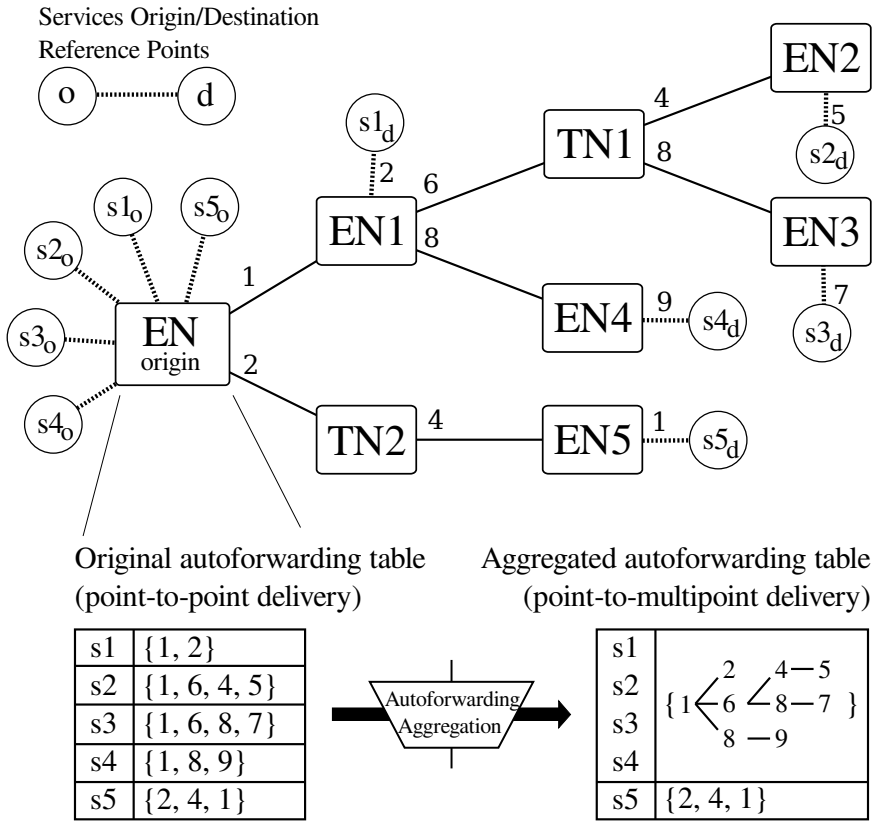


Figure 5.1 Illustration of the point-to-multipoint autoforwarding aggregation function on a small example

very simple to implement and very efficient [1]. What we propose here is to extend autoforwarding to point-to-multipoint transmissions. Hereafter, we call point-to-multipoint autoforwarding the ability to transmit a packet starting from one point towards several distinct destinations by using exclusively the information contained in its initial header.

In our architecture the interest is twice: (i) applied to data packets it makes the architecture able to transport point-to-multipoint client traffics, i.e. to achieve “multicast" (ii) it can also be used for grouping the OAM packets emitted by Edge Node; doing so, it decorrelates the overhead from the amount of services. In order to complement our first proposal (multi-services path discovery), we focus now on the evaluation of the Point-to-multipoint Autoforwarding benefits in the second purpose (OAM aggregation).

1. indeed, TNs have no Forwarding Information Base to store and maintain up to date. As a path is only stored at the EN, no convergence time is involved: a client traffic can be redirected instantly if a better path is found by the routing plane (even frame by frame)

In terms of implementation, the first idea is to use tree data structures instead of lists for the representation of the forwarding ports. The second idea is to build the desired autoforwarding trees on the base of the point-to-point autoforwarding information already present in LOCARN Edge Nodes. Hence no additional protocol is involved for the trees determination (like for example the Protocol Independent Multicast (PIM) protocol suite [17, 24, 35] and Internet Group Management Protocol (IGMP) [28] must be added to make multicast in an IP network). Two symmetric mechanisms must be defined. In ENs an aggregation function must be introduced for the building of Point-to-MultiPoint (P2MP) tables. In TNs and ENs, the autoforwarding function must be extended to process adequately point-to-multipoint header structure on the base of a point-to-multipoint algorithm (P2MP forwarding).

As illustrated in Fig.5.1, the aggregation function is responsible for providing the P2MP tables by using the point-to-point ones. The figure illustrates an example with five services (s1 to s5) terminating on a same Edge Node (called here the “EN origin") and whose destinations are distributed among other ENs. Over time the EN origin routing process fill/update the point-to-point autoforwarding table (on the left). At some points, the aggregation function is performed, filling or updating the P2MP table (on the right) regrouping the paths according to the port similarities. In the example, s1 to s4 are grouped together on a same tree root because they all start their autoforwarding by the port 1, then s2 and s3 paths belong to the same sub-tree branching on the port 6 etc. For the aggregation of OAM packets considered hereafter, we try to regroup all service paths, so we build on table entry, i.e. one tree for each first port which constitutes the tree root. The table entries (autoforwarding header) are encoded according to a tree representation convention in accordance with the autoforwarding algorithm.

The P2MP autoforwarding function is implemented in Transit Nodes and decodes the header information of an incoming packet. It adequately splits into several packets (if needed) before forwarding them (see Algorithm. 1). To do so, we consider the P2MP header encoding as a sequence of *fwdChunks* (2 Bytes) composed of a *fwdPort* (1 Byte) coding the next port as usual, and a *fwdCode* (1 Byte) that indicates how to process the forwarding of the P2MP incoming packet. We respect the following encoding convention: if the *fwdCode* is zero, then the packet is processed as a usual point-to-point packet; on the contrary a *fwdCode* exceeding zero both indicates than the packet must be bisected and indicates where to bisect the P2MP header.

Algorithm 1 Point-to-multipoint Autoforwarding Algorithm

```
1:  procedure AUTOFORWARDINGPROCESSREC(packet p)
2:      header h ← readHeader(p)
3:      integer i ← readFwdHop(h)
4:      integer fwdCode ← readNextFwdCode(h, i)
5:      integer fwdPort ← readNextFwdPort(h i)
6:      if {fwdCode = 0} then
7:          // perform usual point-to-point autoforwarding
8:          setFwdHop(p, i + 1)
9:          sendPacket(p, fwdPort)
10:     else
11:         // fwdCode ∈ [1 − 255] indicates where to bisect
12:         header h1, h2 ← bisectHeader(h, fwdCode)
13:         packet p1 ← assemblePacket(h1, copyPayload(p))
14:         packet p2 ← assemblePacket(h2, copyPayload(p))
15:         sendPacket(p1, fwdPort)
16:         AutoforwardingProcessRec(p2)
17:     end if
18: end procedure
```

The result of the bisection are two packets having two (sub)parts of the initial header, and of course the same payload (that have been copied). The packet based on the first header subpart is ready to be forwarded upon the interface (indicated by *fwdPort*) whereas the P2MP autoforwarding function is relaunched on the second packet whose header may contain several other subheaders. Finally, the recursive P2MP autoforwarding function will be called N times until $fwdCode = 0$ are encountered, where N corresponds to the n-ary tree branching factor of the incomming packet at this node.

The n-ary tree's form resulting from the aggregation function gives us information about the efficiency of path grouping. To estimate this efficiency, the tree density can be used (i.e. the mean internal nodes degree, also called the mean branching factor). But to measure the no redundancy of link usage, we want to take into account the tree balancing by measuring the number of vertices ($|V|$) divided by the tree height: this give us information about both the tree mean density and about its form.

$$TreeAggregationScore = \frac{|V|}{TreeHeight} \tag{5.1}$$

Network	N	L	Mean Density	Diameter
GEANT	22	37	3.4	4
Dtelecom	68	353	10.4	3
Level 3	46	268	11.7	4

Table 5.1 Topological Dimensions of the Simulated Networks

5.2 Performances Evaluation

For the LOCARN simulation, we included networks topologies from the CCNSim simulator [1] in our own *OMNeT++* simulator, whereas the n-ary trees structures implementation relies on the *STLplus* [2] C++ library. Thereafter S designates the amount of services declared within a LOCARN domain whereas N and L are respectively the numbers of nodes and links, and δ_{net} designates the network mean density (i.e. the mean number of neighbors per node).

5.2.1 First Proposal Evaluation

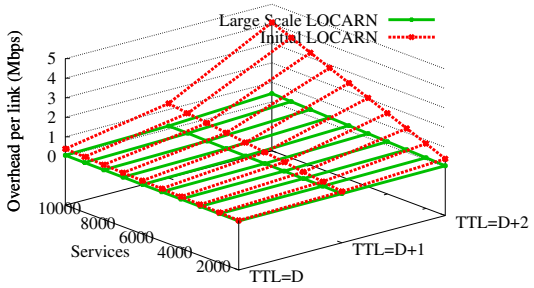
Evaluation Method

Our goal is to estimate the overhead reduction allowed by the multi-services path request proposal compared to the initial design. In both designs, the global overhead generation depends on many parameters related to *the network infrastructure*, *the client layer* (i.e. services) and *the LOCARN's settings*. We proceed as follows.

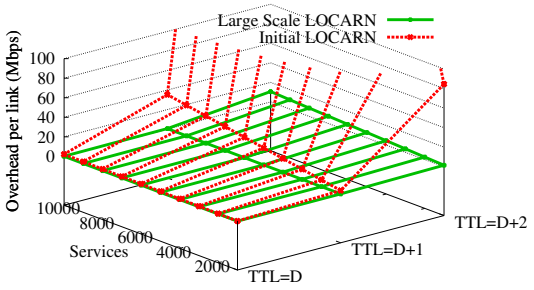
Concerning the network infrastructure, the network topological dimensions (namely the density δ_{net} and diameter D) impact the flooding magnitude [2]. Hence we have selected three well known networks having various topological characteristics, summarized in Table 5.1.

Concerning the client layer, the amount of active services over a domain impacts the overhead linked to the discovery process for both the initial design (each service involves periodical discoveries), and the multi-services based design (packet sizes increase linearly according to the amount of services identifiers to be memorized [3]. Beyond the number of services, the distribution of endpoints within the network is also decisive, what was not the

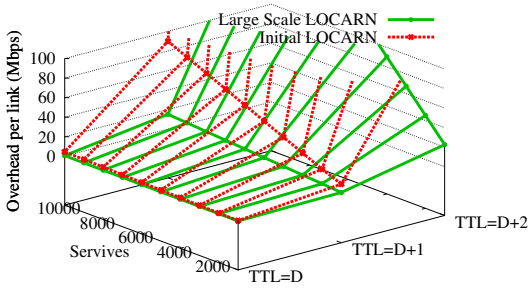
2. the amount of messages generated by a flooding process
3. in the multi-services discoveries, since path requests are grouped by nodes, their amount is bound by the number of nodes rather than the number of services



(a) GEANT Network



(b) DTELECOM Network



(c) LEVEL 3 Network

Figure 5.2 Mean overhead bitrates per link due to floods discoveries: with the initial design besides with the first proposal

case in the initial design[4]. To make no assumption about the traffic matrix, we randomly distribute the services endpoints within the domain, making the number of service reference points per node tending toward S/N when S is high.

Concerning the LOCARN's parameters, first the flooding packet Time To Live (TTL) which is a key factor of the flooding magnitude (see [56]), is set accordingly to the network diameter: with $TTL = D$ any service is able to find at least one path whereas with $TTL > D$ we allow the discovery of longer paths. Generally, a TTL value of $D+2$ is sufficient enough to discover a variety of many paths. Finally, the Optimization Interval (OI) which is the duration between two discovery processes also impacts linearly the amount of discoveries launched per time unit. In the two designs, we fixe $OI = 10s$ in order to assure a good LOCARN routing dynamicity.

4. we observed in the three considered networks that the flood magnitude is almost the same whatever the origin node, since the $TTL > D$

To get the results of the Fig. 5.2, we observe the discovery packets (their amount and their sizes) that have been transmitted during one round of discoveries for all services. By extrapolation, we are then able to give a *mean estimation of the overhead* over time according to the amount of services and the flooding propagation limit (TTL).

Interpretation of Results

Fig. 5.2 exposes as results the mean overheads per link, which permit us to estimate the gain magnitude of the first proposal. Those bitrate values have a very low standard deviation (less than 1%) resulting from data that is quite low (number of packet and packet size). Those values permit to compare the overheads without taking into account possible fluctuations over time [5].

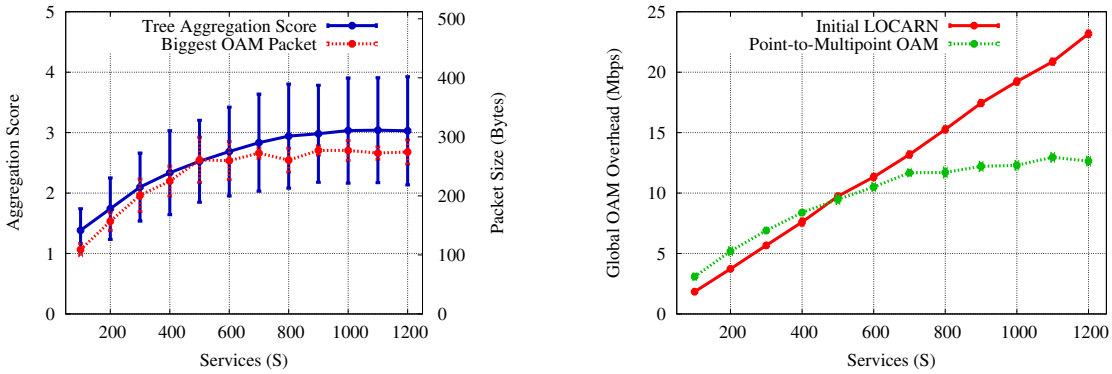
What we observe first in all scenarios is that the amount of services has much more impact on the initial design than on the “Large Scale LOCARN" with the first proposal. In Fig. 5.2a, we can observe that with a network like GEANT (i.e. with a reasonable topological density $\delta_{net} = 3.4$) the interest of our first proposal is globally low (except maybe when both S and TTL are high). In Fig. 5.2b, we observe that the first proposal is very interesting when S increases and $TTL > D$. In Fig. 5.2c, we see that with the first proposal the overhead decorrelation from S is important but not total. With very dense topology like LEVEL 3, the flood magnitude is such important that even with our proposal the overhead becomes too important when both S and TTL are high.

5.2.2 Second Proposal Evaluation

Evaluation Method

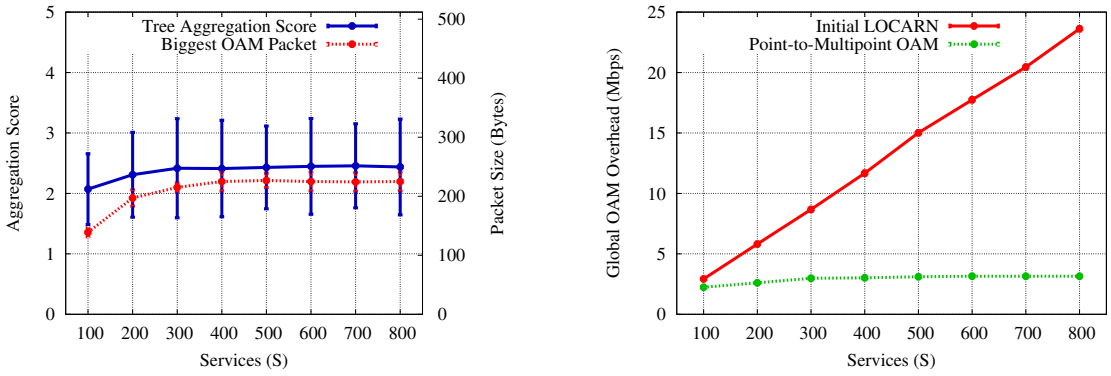
We now estimate the overhead reduction allowed by the application of the point-to-multipoint proposal to LOCARN OAM’s packets transmission. For the two designs, the overhead generated by the OAM packets depends on the number of services and on the service distribution within the network. Hence we assume two modes of services distribution: a “Scarse Mode" where all the reference points (service origin and destination Reference Points) are distributed randomly among the domain; a “Dense Mode" where origins are distributed among an area and destination reference points among another one (areas are composed of contiguous node subsets of about one third of the network).

5. can are mostly significant in the initial design because numerous discoveries can occur among very short durations



(a) Tree Aggregation (b) Overhead Estimation

Figure 5.3 Evaluation of the second proposal on the GEANT Network: Scarse Mode



(a) Tree Aggregation (b) Overhead Estimation

Figure 5.4 Evaluation of the second proposal on the GEANT Network: Dense Mode

In Fig. 5.3 and Fig. 5.4, the left and right diagrams give the trees aggregation scores and the global OAM overhead comparisons between the initial design and the second proposal, withing repectively a “scarse" and a "dense" services distribution over the GEANT topology. The aggregation scores of left figures (solid line, read on left axis) are estimated with the formula 5.1 averaged among all trees resulting from the aggregations. We also evaluate the maximum sizes of the OAM packet generated (dotted line, read on the right axis), which are related to trees height [6]. The global overheads represented in right figures are obtained by extrapolation of simulation results. After all the services paths have been found, we send one round of OAM packets: for the initial design we simply build packets with the usual point-to-point autoforwarding table and send one packet for each service whereas for the point-to-multipoint version, all Edge Nodes launch their aggregation functions to build their

6. the biggest packets are ones that are sent on the first link and that have not been splited yet

point-to-multipoint autoforwarding tables and then send one “big" packet for each output port. The size of the OAM packets is observed along their end-to-end forwarding . On the base of the cumulated sizes obtained for one round with S services in each design, we estimate the global overheads by considering the Service Check Interval (SCI) equal to 10ms – a such period makes the architecture able to reach a sub-50ms service protection in most network topologies.

Interpretation of Results

As might be expected, the point-to-multipoint OAM design becomes more gainful from a certain amount of services (visible in Fig. 5.3b) which is due to the introduction of *fwdCode* that is not efficient if the level of aggregation is too low.

Then when S increases, in so far the routing policy leads to spreads the end-to-end paths selected, the level of paths aggregation increases tendentiously. At a certain point, the aggregation score reaches a maximum because no distinct paths are found anymore, trees are somehow “saturated". This makes the gain even more radical because the increase of service has no impact on the packet size anymore. The aggregation saturation is both observable with the *TreeAggregationScore* (Fig. 5.3a and 5.4a) and the global overheads (Fig. 5.3b and 5.4b). As might be expected, saturation is faster in a Dense than in a Scarse distribution whereas the potential of aggregation is lesser. Finally, we see that on the GEANT network example, the second proposal usage for the OAM overhead minimization would quickly be interesting when the number of services exceeds one thousand.

In terms of performance, the P2MP autoforwarding function based on the Algorithm 1 involves an amount of operation per packet related to the number of tree’s branches at the considered step. Hence, the *TreeAggregationScore* values permit us to estimate the number of recursive calls per packet and per node: results are widely acceptable in our example. On the other hand, we can observe that the biggest size for OAM packets, which also depends on the *TreeAggregationScore*, does not become tremendous on our evaluation.

5.3 Perspectives of the second proposal

The second proposal allows to completely meet the connection types defined by the ITU-T recommendations: (i) Uni-directional point-to-point connection ; (ii) Bidirectional point-to-point connection (iii) *Uni-directional point-to-multipoint connection*. Indeed, with point-

to-multipoint autoforwarding, LOCARN is intrinsically able achieve point-to-mulipoint transportation without the addition of any control protocol both for client subscription (i.e. IGMP [28]) or tree management (i.e. PIM [17, 24, 35]) because P2MP trees are built in source nodes according to the possible agregation of end-to-end paths at each moment, it is to say opportunistically. Here again, the question of service identifiers unicity or groups assignation and management is left to upper layers : the LOCARN's role is limited to the establishment of communications between the declared access points.

Chapter 6

Conclusion

6.1 Concluding Remarks

LOCARN is a simple, adaptive, resilient and plug-and-play packet architecture which is composed by three network mechanisms: *autoforwarding*, *enhanced flooding* and *end-to-end fault detection*. Because of its properties, it can be envisioned in many different contexts for the quick network deployment and the quick establishment of adaptive and resilient point-to-point communications without involving configuration. In this thesis, we focused on LOCARN as an alternative Packet Based Transport Network (PTN) technology for operators. In such a perspective, the architecture's principale breakthrough is its radical design simplicity and its highly opportunist adaptiveness if compared to the transport network solutions for now. Since those properties come with a significant control plane overhead production, this counterpart has been the critical point to study (chapter 4) and to improve (chapter 5).

Retrospectively, the core contribution of this thesis is to bring out that under the consideration of high datarates links ($> 1Gbps$), the overhead cost is widely acceptable, and even negligible in realistic backbone use cases. Indeed, the MPLS standards were built on old assumptions about IP infrastrutures capabilities of transmissions. Nowadays, considering the constant grow of bandwidth capacities provided by optical transmissions the LOCARN radical approache based on floodings becomes more and more interesting.

Concretely, LOCARN can be retained as an anticipated answer to future network operators' needs for high responsiveness to clients demands without the involvement of any preliminary resource reservation process. Typical evolutions that suggest those needs are

the expension of the Cloud and the works in the Big Data areas. Beyond those trends, LOCARN constitutes a mean for operators to reach a more flexible business model for the transport of information.

6.2 Perspectives

In-depth studies

The intra-DC use case and energy saving The context of application can be orinted towards tha Data Center Use Case. Indeed, its simplicity and adaptiveness make LOCARN a suitable intra-DC solution. Moreover, the LOCARN self-adaptiveness could be coupled with mechanism(s) to turn on / turn off network devices, in order to solely keep alive the needed devices, and thus minimize energy consumption.

Study of LOCARN paths adapting behaviors In order to confort the interest of LOCARN usage though it differents Use Cases, the evolution of services paths distribution over time could be studied. Such studies would be done by following concrete examples of network topologies and their associated traffic matrix whereas the varying parameters would be the client traffics characteristics of evolution and the LOCARN parameters including services' path selection rules. Such studies would allow to find the suitable LOCARN parameters to associate to a specific context of use.

Point-to-MultiPoint (P2MP) data plane experimentation Initialy the architecture is able to provides point-to-point transport services ; it can also easily be extended to point-to-multipoint transport as presented in chapter 5. Yet, to propose it for transport networks, point-to-multipoint autoforwarding computation cost should be quantified for very high traffics to other packet forwarding processes, preferably by experimentation.

Quality Of Service Extensions LOCARN is conceived as a best effort Packet Transport Architecture. Nevertheless, some packet priorization mechanisms could be easily introduced in order to cope with various clients needs in terms of QoS.

Implementation The most promizing LOCARN implementation perspective at the time is the integration to a SDN implementation, it is to say to “program" the LOCARN behaviors

in a SDN framework. For now, this is not feasible under the current openflow specifications [1] yet it should be later.

1. do not allow packets headers to be modifed on the fly, which is necessary for the LOCARN autoforwarding data plane

Publications from this thesis

Publications with reading commitee:

- International conference –
  This work received the "Best Paper Award" prize of the conference
  D. Le Quere, C. Betoule, R. Clavier, Y. Hadjadj-Aoul, A Ksentini, and G. Thouenon. Scalability & performances evaluation of LOCARN: Low opex and capex architecture for resilient networks. In Innovations for Community Services (I4CS), 2014 14th IEEE International Conference on, pages 1–8, June 2014
- Journal –
  D Le Quere, C Betoule, R. Clavier, Y Hadjadj-Aoul, A. Ksentini, and G. Thouenon. Presentation & evaluation of LOCARN: Low opex and capex architecture for resilient networks. Studia Informatica Universalis, 12, June 2014.
- International workshop –
  D. Le Quere, C. Betoule, R. Clavier, Y. Hadjadj-Aoul, A Ksentini, and G. Thouenon. Towards a large scale LOCARN design. In Globecom, 2014 IEEE International Conference, December 2014.

Pending paper:

- C. Betoule, G. Thouenon, D. Le Quere and M. Salaün. LOCARN: An innovative Plug&Play auto-adaptative packet transport network.

Other publication:

- *Poster Orange "Journée des doctorants"* D. Le Quere. An innovative Plug&Play auto-adaptative packet transport network. September 2012.

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Appendix A

LOCARN Modeling

A.1 LOCARN Services: State Machine

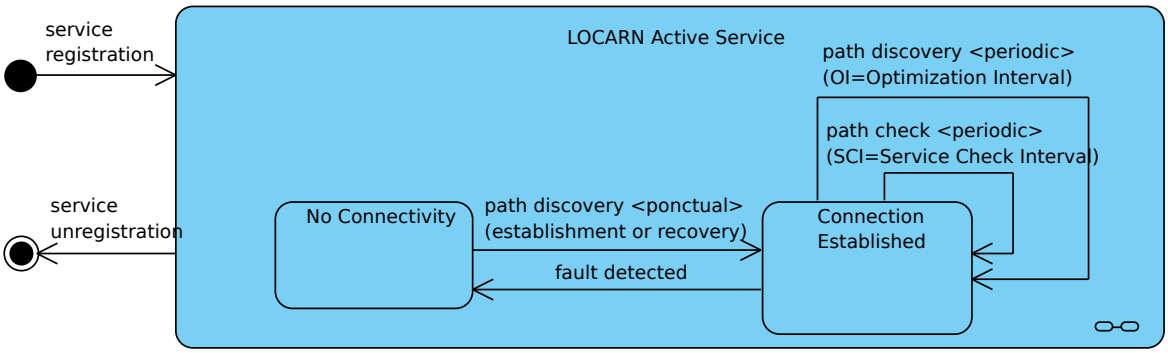


Figure A.1 Service Machine State Diagram (Control Unit Process)

A.2 Sequences Charts: Initial LOCARN Design

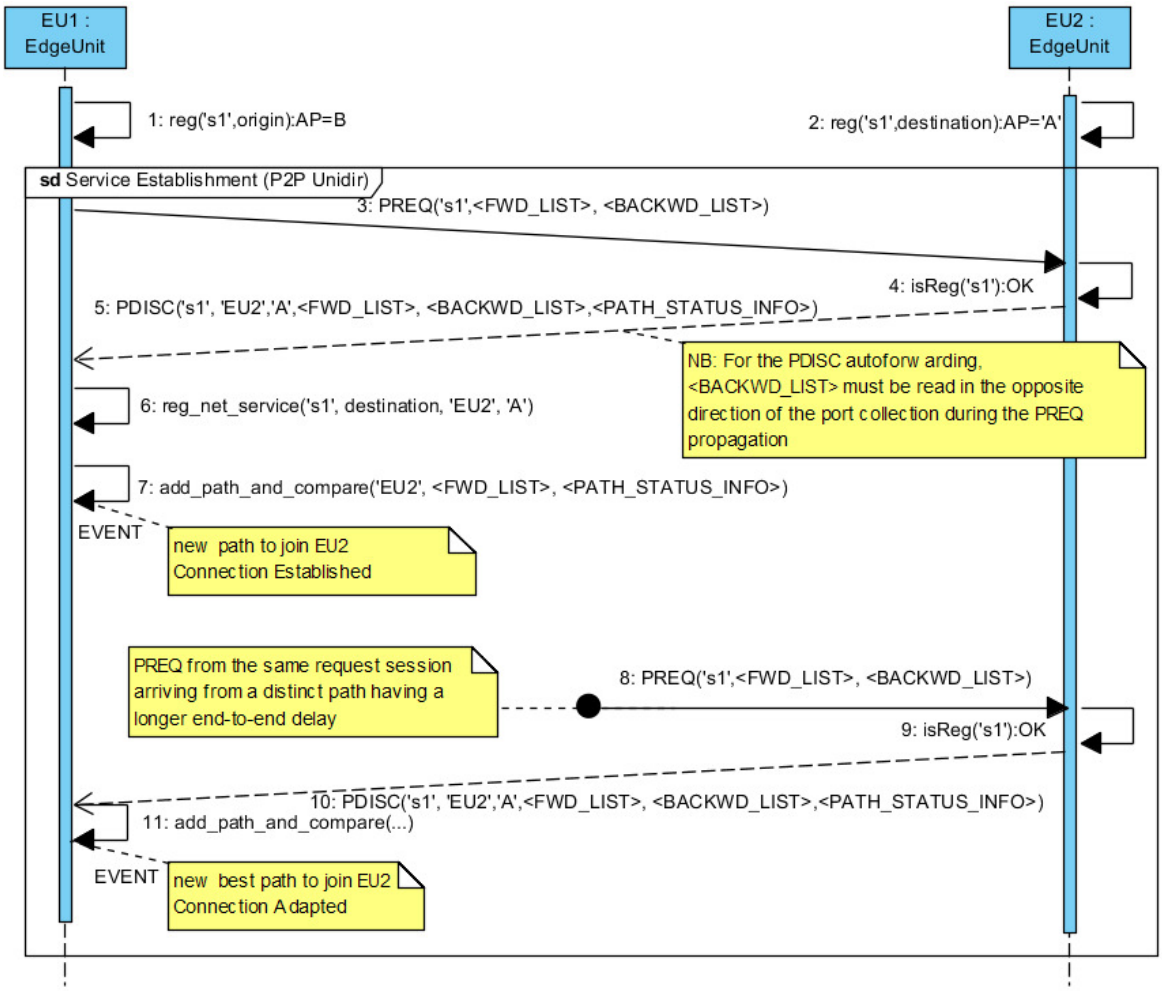


Figure A.2 P2P Service Establishment and Adaptation (Unidir)

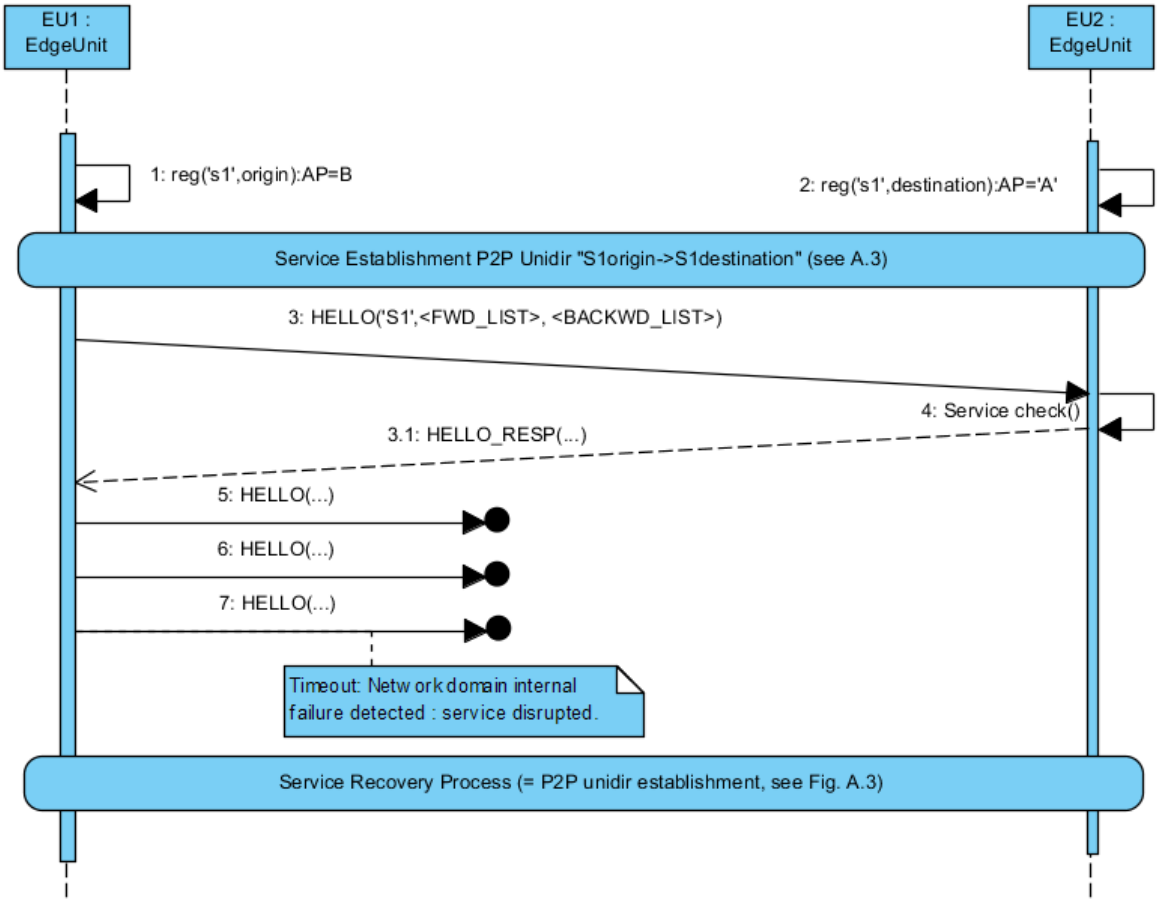


Figure A.3 P2P Service Supervision and Internal Fault Detection (Unidir)

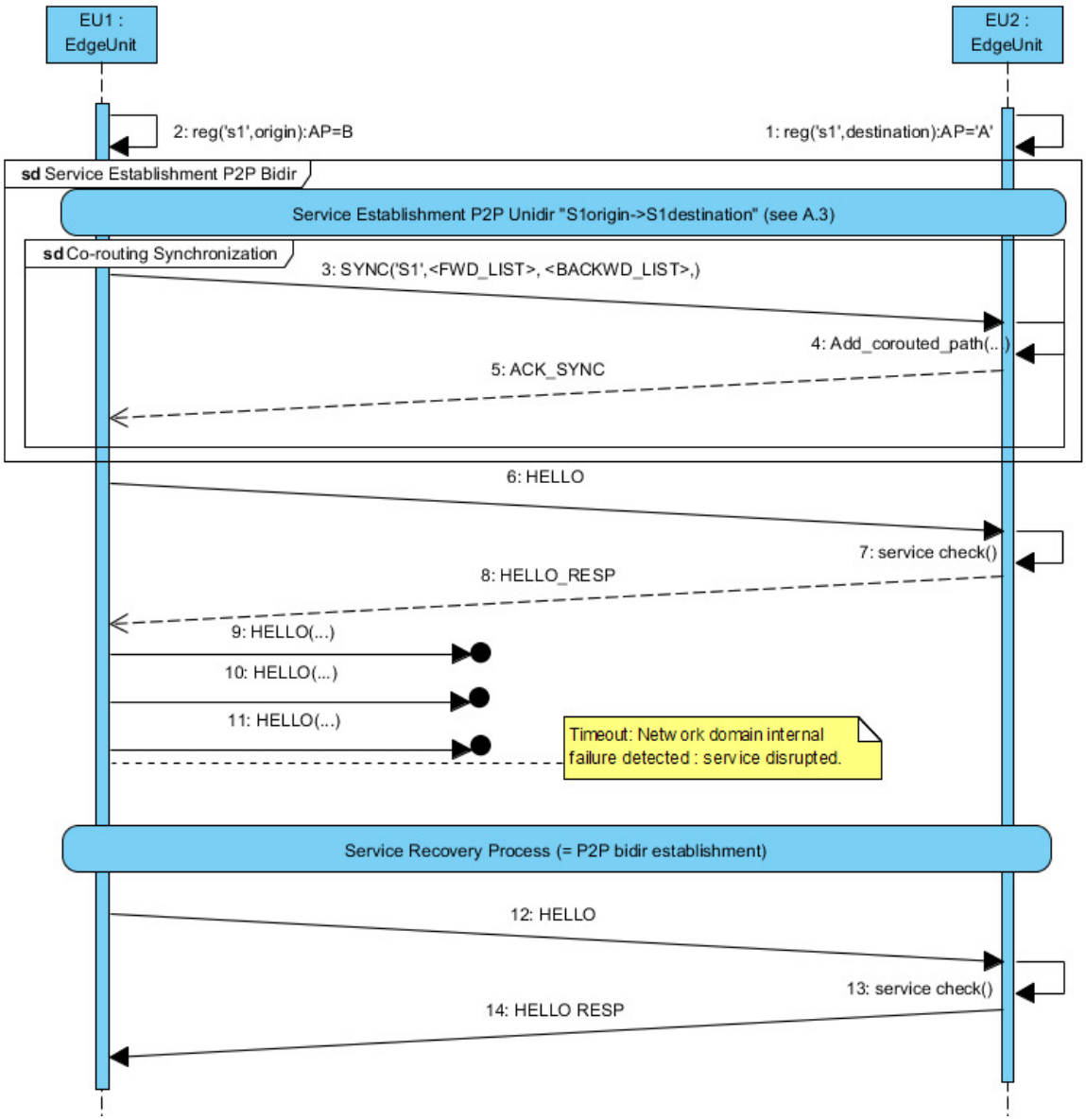


Figure A.4 P2P Service Supervision and Corouting Synchronization (Bidir)

A.3 LOCARN Simulator Overview

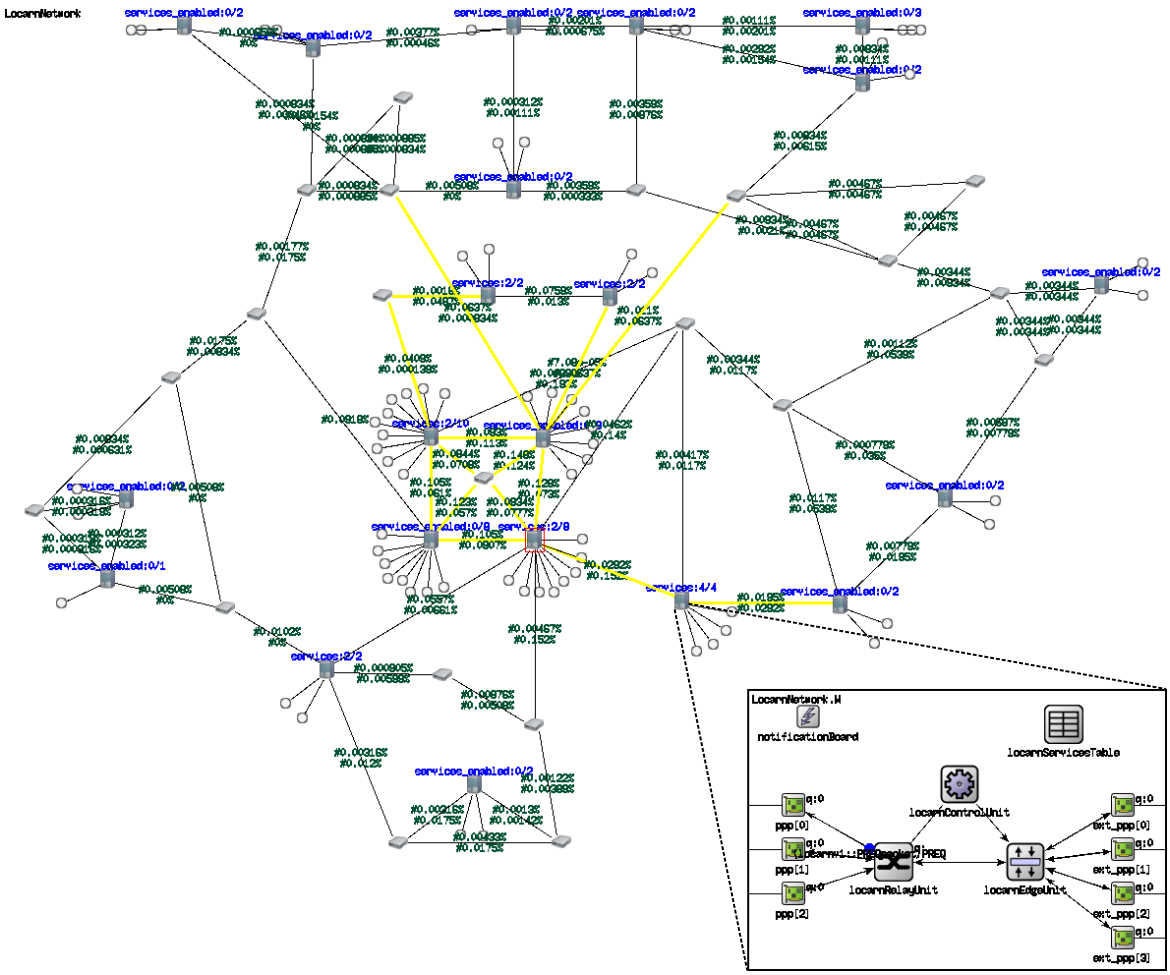


Figure A.5 A view of the LOCARN OMNeT++ implementation upon a backbone network topology and services distribution (phase of flooding)

Conception et étude des performances d'une solution auto-configurable pour les réseaux de transport du future

Dans cette thèse, nous étudions LOCARN : « Low Opex & Capex Architecture for Resilient Networks ». LOCARN est une architecture de réseaux paquet alternative conçue pour être la plus simple possible. Simple dans sa structure fonctionnelle d'une part, avec l'implication d'un petit nombre de mécanismes et de composants réseaux. Simple dans sa gestion opérationnelle d'autre part, avec la réduction au minimum de l'effort de déploiement et de maintenance pour le gestionnaire du réseau. Compte tenu de la complexification croissante des réseaux de transport opérateurs ces dernières décennies, nous considérons ici ces réseaux comme le cas d'application privilégié. Dans le contexte des réseaux de transport, LOCARN constitue une opportunité considérable de simplification par rapport aux technologies actuelles en permettant à la fois des réductions de CAPEX[1] et d'OPEX[2].

Présentation générale de l'architecture LOCARN

Les diverses fonctions qui composent un réseau de télécommunication peuvent être classées en deux groupes fonctionnels principaux. Le premier est le groupe fonctionnel de transport qui couvre le transfert de toutes les informations de télécommunication d'un point à un autre. L'autre, le groupe fonctionnel de commande, gère divers systèmes et services auxiliaires ainsi que les fonctions de maintenance. Du point de vue du groupe fonctionnel de transport, LOCARN est une couche réseau établissant des communications paquets point-à-point à travers un domaine réseau défini pour une couche « cliente » (qui peut être une autre couche réseau ou bien une couche applicative) et s'appuyant sur une couche réseaux « serveur ». Nous qualifions la solution LOCARN d'architecture dans la mesure où celle-ci définie à la fois un plan de transport et un plan de contrôle et intégrant dans sa conception les interactions entre ces deux plans. Le service de transport paquet offert aux clients est basé sur un mécanisme d'*auto-acheminement des paquets*. Les communications point-à-point sont établies et maintenues dans le plan de transfert par le plan de contrôle via des mécanismes de routage et maintenance dans le nœud source. Ces mécanismes sont basés sur une *inondation intelligente du réseau*. Cette

[1] CAPEX : « Capital Expenditure »
[2] OPEX : « Operational Expenditure »

conception originale permet à l'architecture d'avoir des propriétés de résilience et d'auto-adaptation en toute simplicité.

Présentation technique de l'architecture LOCARN

1/ Un plan de transfert basé sur des paquets auto-acheminés – Notre architecture utilise des paquets pour transporter l'information cliente, elle définit à cette fin un datagramme spécifique pour les paquets de données. Comme pour n'importe quel réseau paquet, les paquets de données sont acheminés d'un point à un autre du réseau par des processus d'aiguillage successifs réalisés au niveau de chaque nœud – un processus d'aiguillage détermine l'interface de sortie du paquet, ou « prochain saut ». Dans LOCARN, les paquets sont dits « auto-acheminés » dans la mesure où le prochain saut d'un paquet est uniquement basé sur des informations contenus dans le paquet lui-même. Le nœud d'entrée du domaine, en charge de l'encapsulation, va intégrer au paquet les informations du chemin à suivre jusqu'à la destination, typiquement une liste d'interface de sortie à emprunter. Ainsi ce plan de transfert est à la fois déterministe et « sans état » dans les nœuds intermédiaires, contrairement aux autres réseaux paquets impliquant la mémorisation de tables de transferts.

2/ Un plan de contrôle basé sur des inondations intelligentes du réseau – Notre architecture est basée sur une inondation intelligente du réseau pour la découverte des chemins. Le découverte/sélection d'un chemin est propre à chaque « service » (communication cliente à établir). Pour établir une communication point-à-point, un des deux nœuds est considéré comme source de la communication et va déclencher l'inondation du réseau avec des paquets spécifiques de découverte, mémorisant chacun leur chemin parcouru. Le nœud considéré comme destination va alors recevoir un ensemble de paquets de découvertes correspondant à des chemins potentiels depuis la source. En répondant à chaque paquet de découverte (à l'aide d'un autre type de paquet), il va permettre à la source de récupérer un ensemble de chemins disponibles et distincts les un des autres. Il lui suffira ensuite de choisir le meilleur d'entre eux. Les chemins sont connus au minimum comme une liste d'interfaces de sortie, mais des informations supplémentaires sur les liens parcourus peuvent être récupérées (délai moyen, bande passante disponible). La communication est établie instantanément dès lors qu'un chemin est sélectionné par le nœud source. En effet, nul besoin de la signaler à aucun autre nœud l'établissement de la connexion : celle-ci est établie dès que le nœud source est capable d'intégrer les informations d'auto-acheminement aux paquets client entrants. LOCARN fonctionne donc suivant un routage dit « par la source » qui n'implique aucune complication,

tant au niveau algorithmique : la découverte remplace le calcul ; qu'au niveau de la mémoire : un nœud ne mémorise que des chemins réellement utilisés et dont il est la source[3].

Par ailleurs ce type de routage autorise la sélection de chemins élastiques (pas nécessairement les plus courts en nombre de sauts) permettant ainsi des adaptations judicieuses du plan de transfert en fonction de l'évolution du réseau. En effet, selon les règles de sélection des meilleurs chemins l'architecture est capable d'adapter ses chemins de façon opportuniste pour optimiser les performances selon les priorités du routage : délais de bout en bout ou répartition de la bande passante par exemple.

3/ La maintenance des communications assurée de bout en bout – Pour obtenir un niveau élevé de résilience des communications, l'architecture prévoie une fois que ces dernières sont établies, des échanges très fréquents de (petits) paquets spécifiques pour la maintenance. Ces échanges permettent d'attester de la viabilité d'une connexion à chaque instant. Au bout d'un intervalle de temps correspondant à trois paquets non reçus, un nœud source va considérer une connexion comme perdue et va relancer un processus de routage pour la restaurer (inondation et sélection d'un chemin). En fonctionnant ainsi, on est capable de rétablir une communication dans des délais très courts qui permettent largement d'atteindre les exigences de disponibilité traditionnelles des réseaux de transport (en moins de 50ms). Par ailleurs les paquets de maintenance sont acheminés selon le même processus que les paquets de données, permettant de conserver un plan de transfert très simple. La résilience de l'architecture est ainsi basée sur la réactivité plutôt que sur la configuration (complexe) de mécanismes de protections utilisées dans les autres technologies de transports et basée sur la détermination d'un chemin/lien de secours de façon proactive.

Contenu du manuscrit

Dans le présent manuscrit, nous étudions dans un premier chapitre les réseaux de transport opérateurs en nous appuyant sur les standards afin de comprendre les évolutions historiques et afin de positionner LOCARN de façon pertinente par rapport aux différentes tentatives d'automatisation de ces derniers au cours du temps. Dans un second chapitre, nous présentons notre architecture de façon détaillée en mettant en évidence ses points forts pour les opérateurs, les défis techniques à relever et en la comparant à des solutions conceptuellement proches comme le protocole Dynamic Source Routing (DSR) ou l'architecture APLASIA. Nous

[3] Contrairement aux protocoles de routage traditionnels de l'Internet, à savoir IS-IS et OSPF

remarquons dans cette partie que la capacité de mise à l'échelle de LOCARN pour des réseaux de grandes dimensions est la question majeure dans notre cas d'application. Pour répondre à cette question, nous évaluons dans le chapitre suivant les sur-débits liés aux paquets produits par le plan de contrôle au moyen d'analyse, de simulation et de models statistiques. Le but général de ces travaux étant de nous assurer que les sur-débits ne rendent pas l'architecture inefficace. Ces études nous permettent d'établir que LOCARN est capable de passer à l'échelle pour des réseaux de transport de dimensions réalistes. Pour le cas d'usage réaliste d'un réseau national, nous obtenons en particulier des estimations de sur-débits systématiquement inférieures à 2% des capacités en bande passante.

Enfin, pour améliorer les performances de l'architecture, nous avons spécifié et évalué deux améliorations de conception pour gérer de façon efficace un très grand nombre de clients. Les résultats obtenus par simulation sur des réseaux bien connus comme GEANT et Deutsch Telecom sont très encourageants. Par ailleurs, la deuxième amélioration de conception nous permet d'établir et maintenir des communications point-à-multipoint sans nécessiter aucun protocole de multicast supplémentaire. La conclusion de ces travaux est que l'architecture finale ainsi proposée constitue une alternative viable pour l'émergence de réseaux de transport paquets simple et auto-adaptatifs.